Deployments, Jobs, and Scaling

Learn how to ...

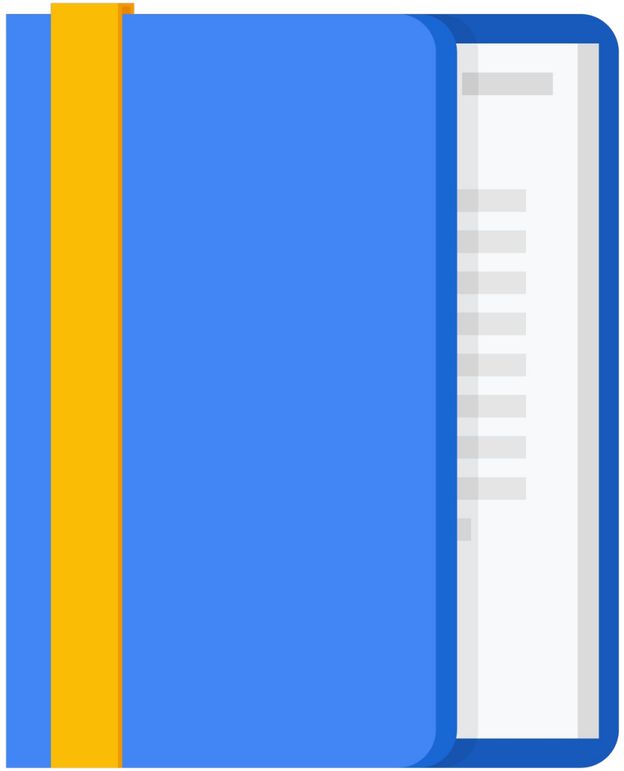
Create and use Deployments.

Create and run Jobs and CronJobs.

Scale clusters manually and automatically.

Configure Node and Pod affinity.

Get software into your cluster.



Agenda

Deployments

Lab: Creating Google Kubernetes Engine Deployments

Jobs and CronJobs

Lab: Deploying Jobs on Google Kubernetes Engine

Cluster Scaling

Controlling Pod Placement

Getting Software into Your Cluster

Lab: Configuring Pod Autoscaling and Node Pools

Summary

Deployments declare the state of Pods

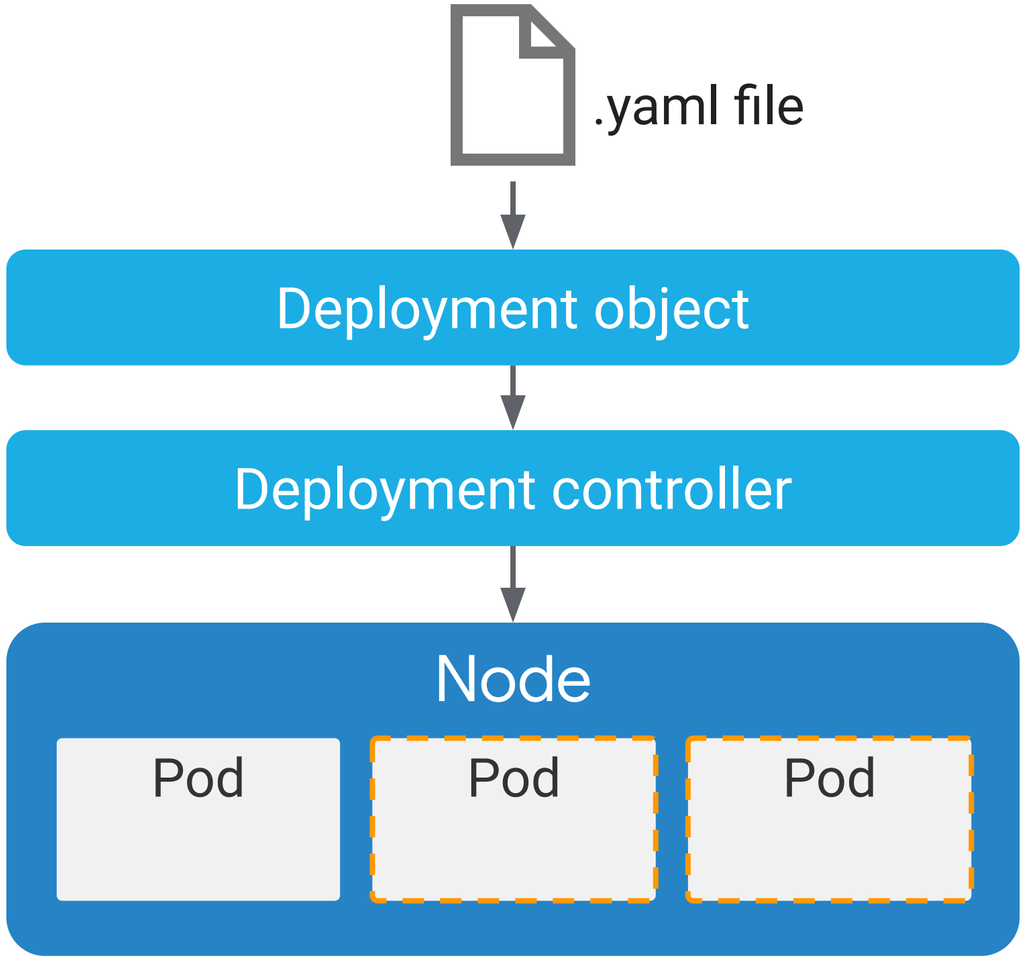
Roll out updates to the Pods

Roll back Pods to previous revision

Scale or autoscale Pods

Well-suited for stateless applications

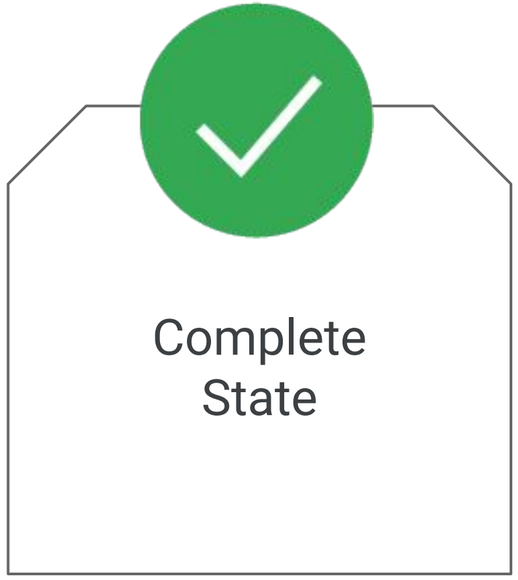
Deployment is a two-part process

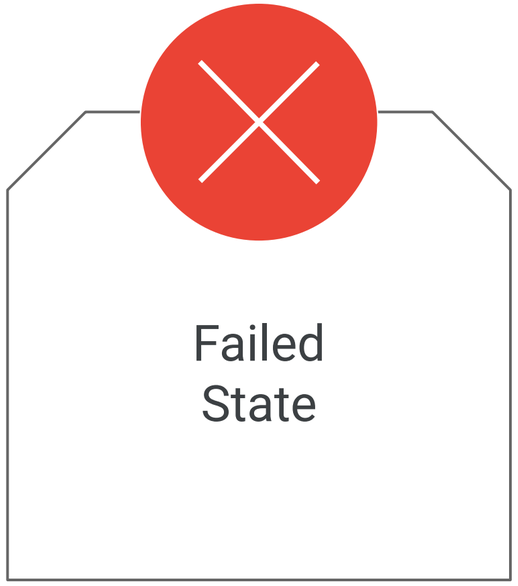


Deployment object file in YAML format

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
      - name: my-app
        image: gcr.io/demo/my-app:1.0
        ports:
        - containerPort: 8080
```

Deployment has three different lifecycle states





There are three ways to create a Deployment

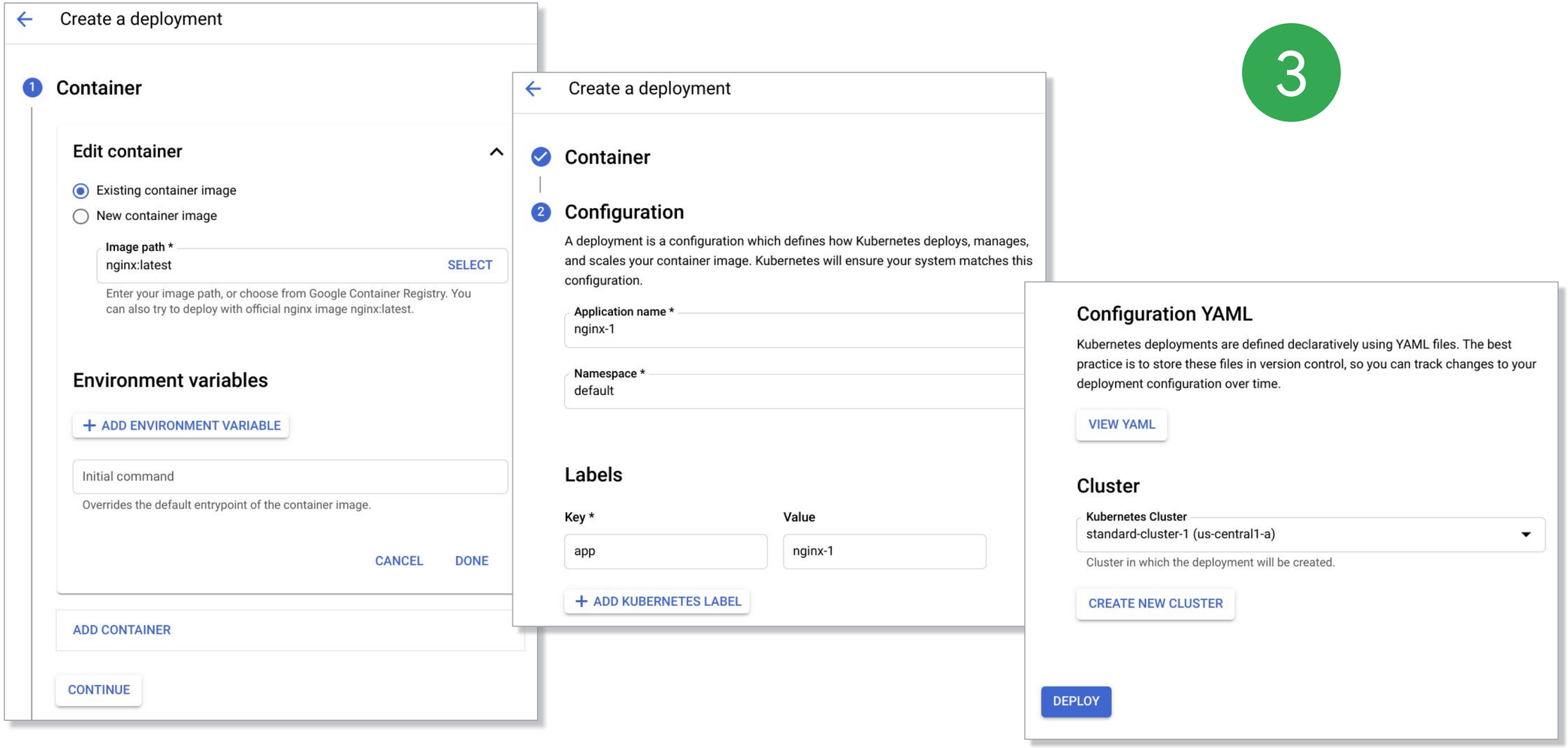
1

```
$ kubectl apply -f [DEPLOYMENT_FILE]
```

2

```
$ kubectl create deployment \
[DEPLOYMENT_NAME] \
  --image [IMAGE]:[TAG] \
  --replicas 3 \
  --labels [KEY]=[VALUE] \
  --port 8080 \
  --generator deployment/apps.v1 \
  --save-config
```

There are three ways to create a Deployment



Use kubectl to inspect your Deployment, or output the Deployment config in a YAML format

```
$ kubectl get deployment [DEPLOYMENT_NAME]
```

```
master $ kubectl get deployment nginx-deployment
NAME               DESIRED   CURRENT   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3         3         3            3           3m
```

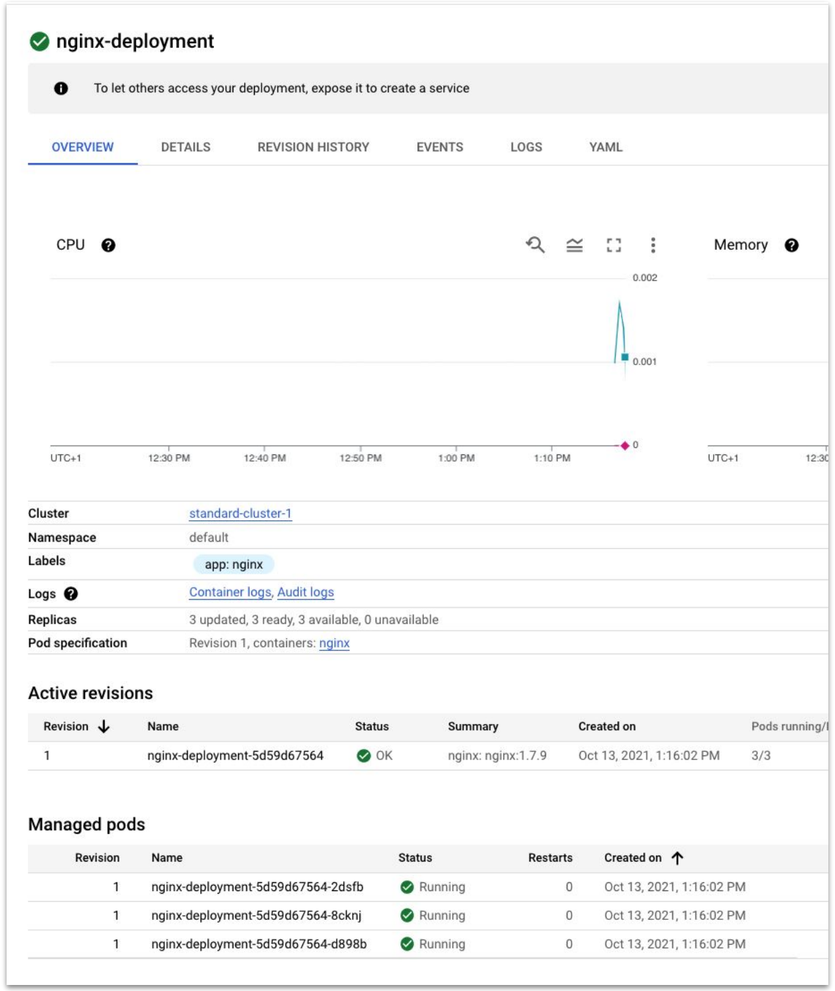
```
$ kubectl get deployment [DEPLOYMENT_NAME] -o yaml > this.yaml
```

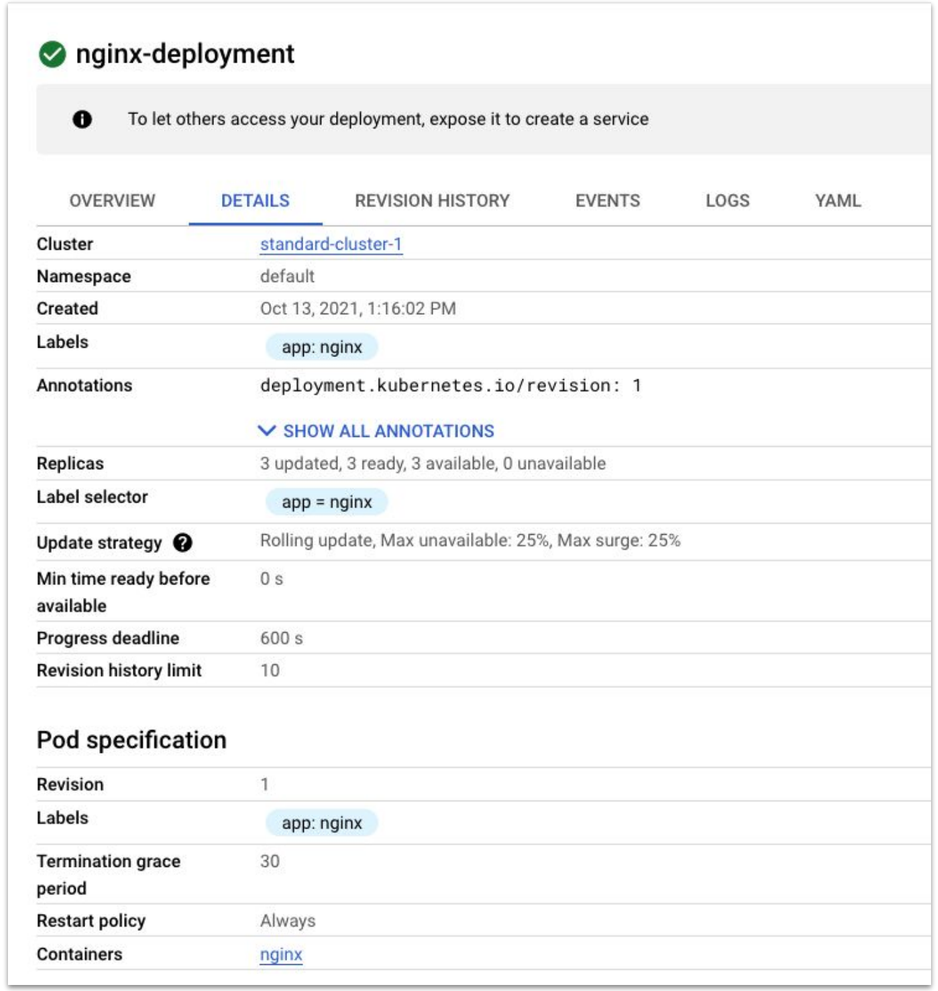
Use the ‘describe’ command to get detailed info

```
$ kubectl describe deployment [DEPLOYMENT_NAME]
```

```
master $ kubectl describe deployment nginx-deployment
Name:                   nginx-deployment
Namespace:              default
CreationTimestamp:      Fri, 12 Oct 2018 15:23:46 +0000
Labels:                 app=nginx
Annotations:            deployment.kubernetes.io/revision=1
Selector:               app=nginx
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:        nginx:1.15.4
    Port:         80/TCP
    Host Port:    0/TCP
```

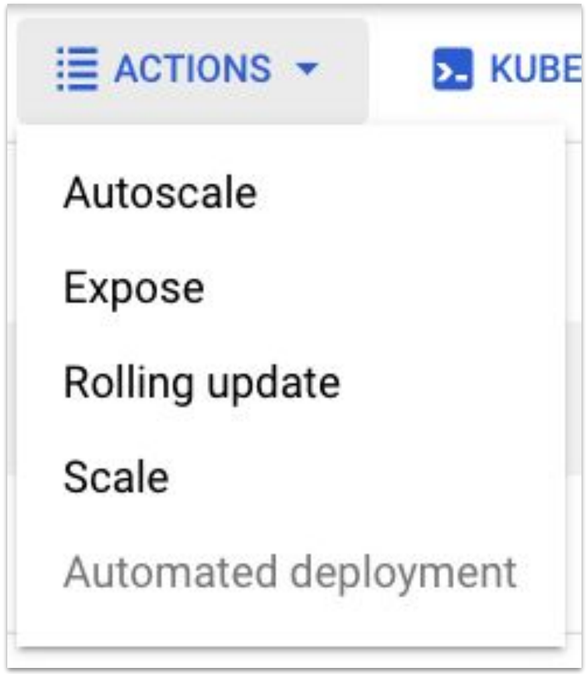
Or use the Cloud Console

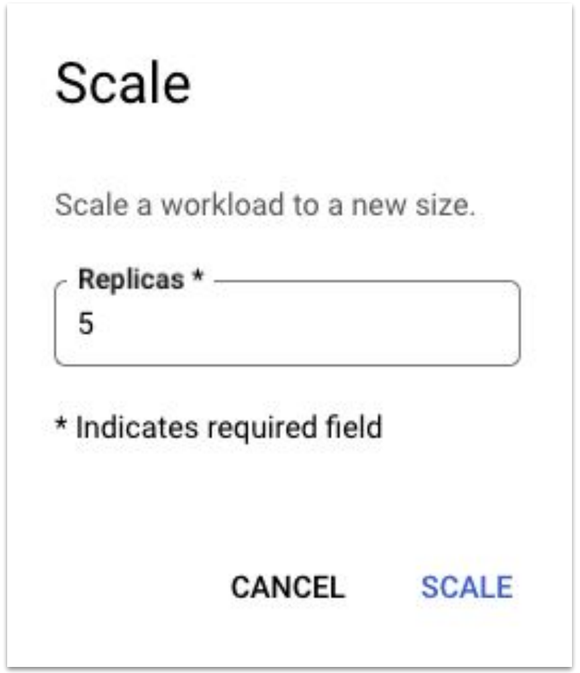




You can scale the Deployment manually

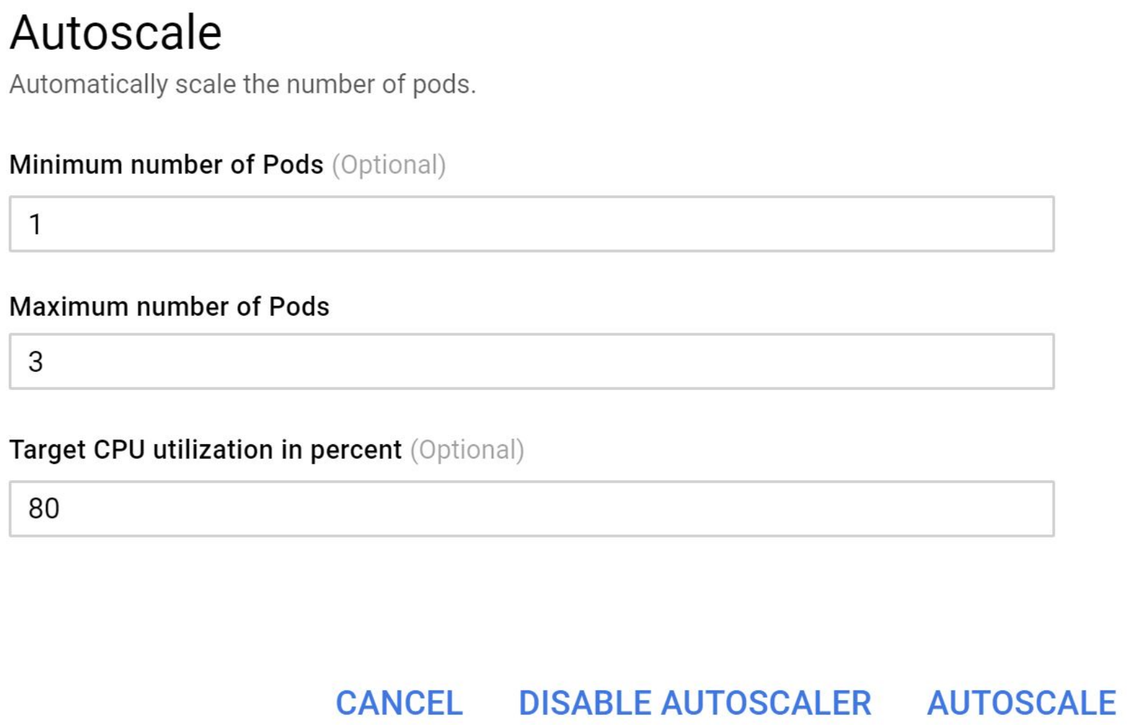
```
$ kubectl scale deployment [DEPLOYMENT_NAME] -replicas=5
```





You can also autoscale the Deployment

```
$ kubectl autoscale deployment [DEPLOYMENT_NAME] --min=5 --max=15
--cpu-percent=75
```



Thrashing is a problem with any type of autoscaling

Cooldown/delay support:

```
--horizontal-pod-autoscaler-downscale-stabilization
```

You can update a Deployment in different ways

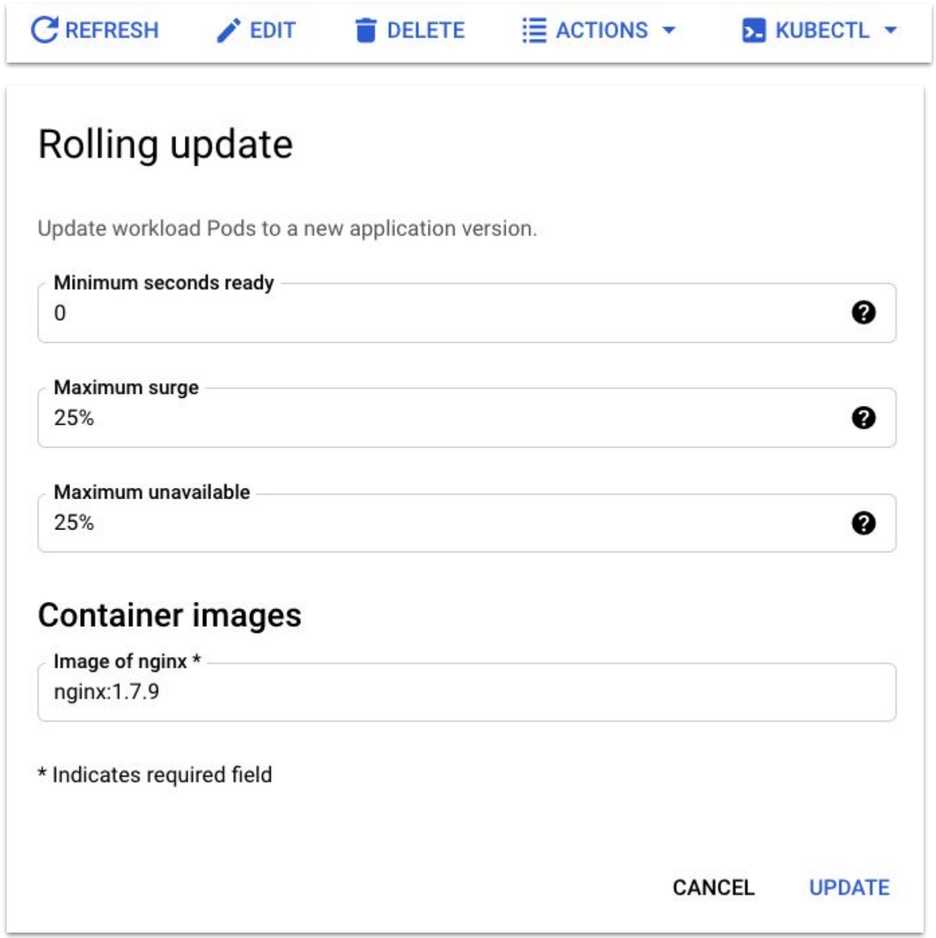
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3
  template:
    spec:
      containers:
      - name: my-app
        image: gcr.io/demo/my-app:1.0
        ports:
        - containerPort: 8080
```

```
$ kubectl apply -f [DEPLOYMENT_FILE]
```

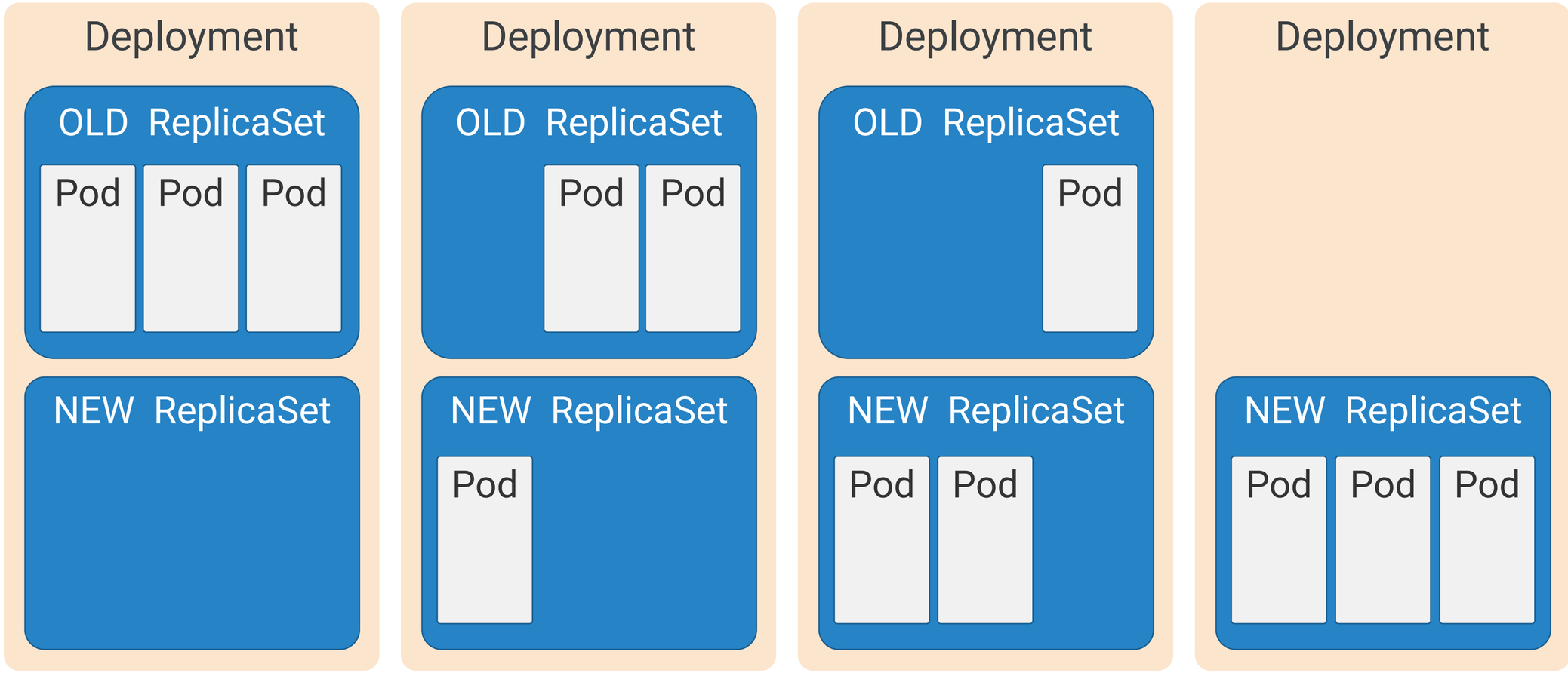
```
$ kubectl set image deployment
[DEPLOYMENT_NAME] [IMAGE] [IMAGE]:[TAG]
```

```
$ kubectl edit \
   deployment/[DEPLOYMENT_NAME]
```

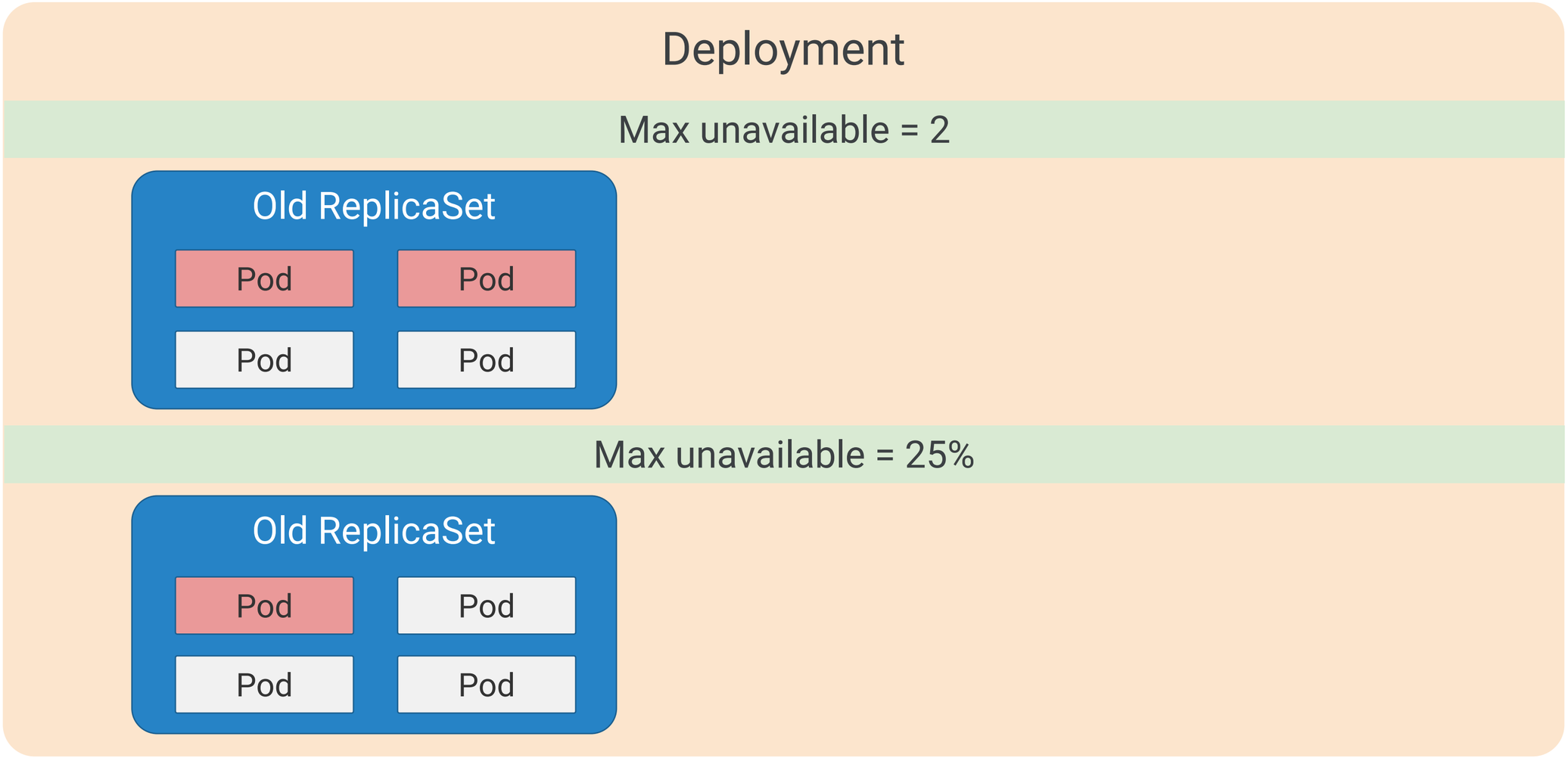
You can update a Deployment in different ways



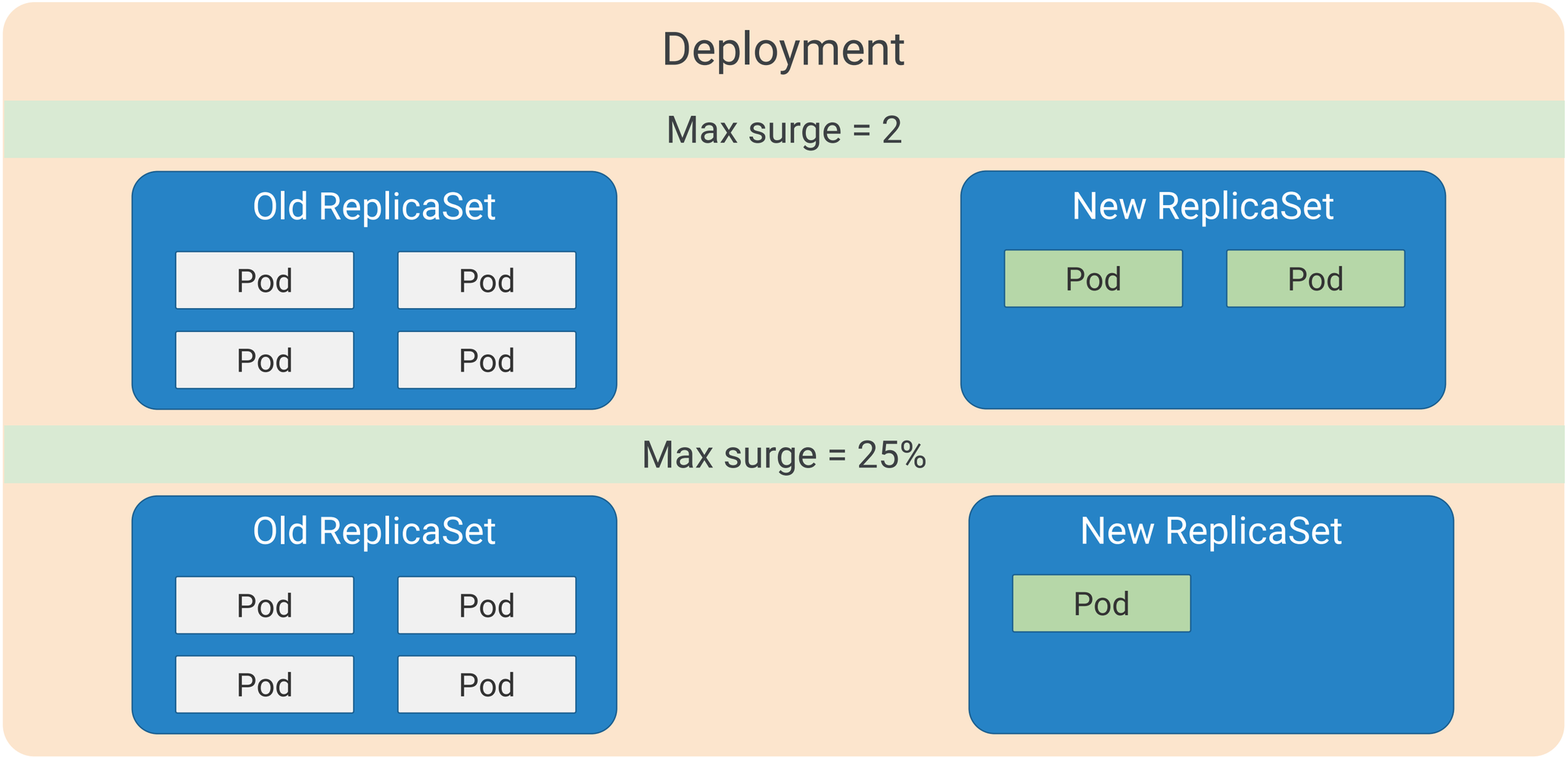
The process behind updating a Deployment



Set parameters for your rolling update strategy



Set parameters for your rolling update strategy



An example of a rolling update strategy

```
[...]
kind: deployment
spec:
  replicas: 10
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 5
      maxUnavailable: 10%
[...]
```

An example of a rolling update strategy

Deployment

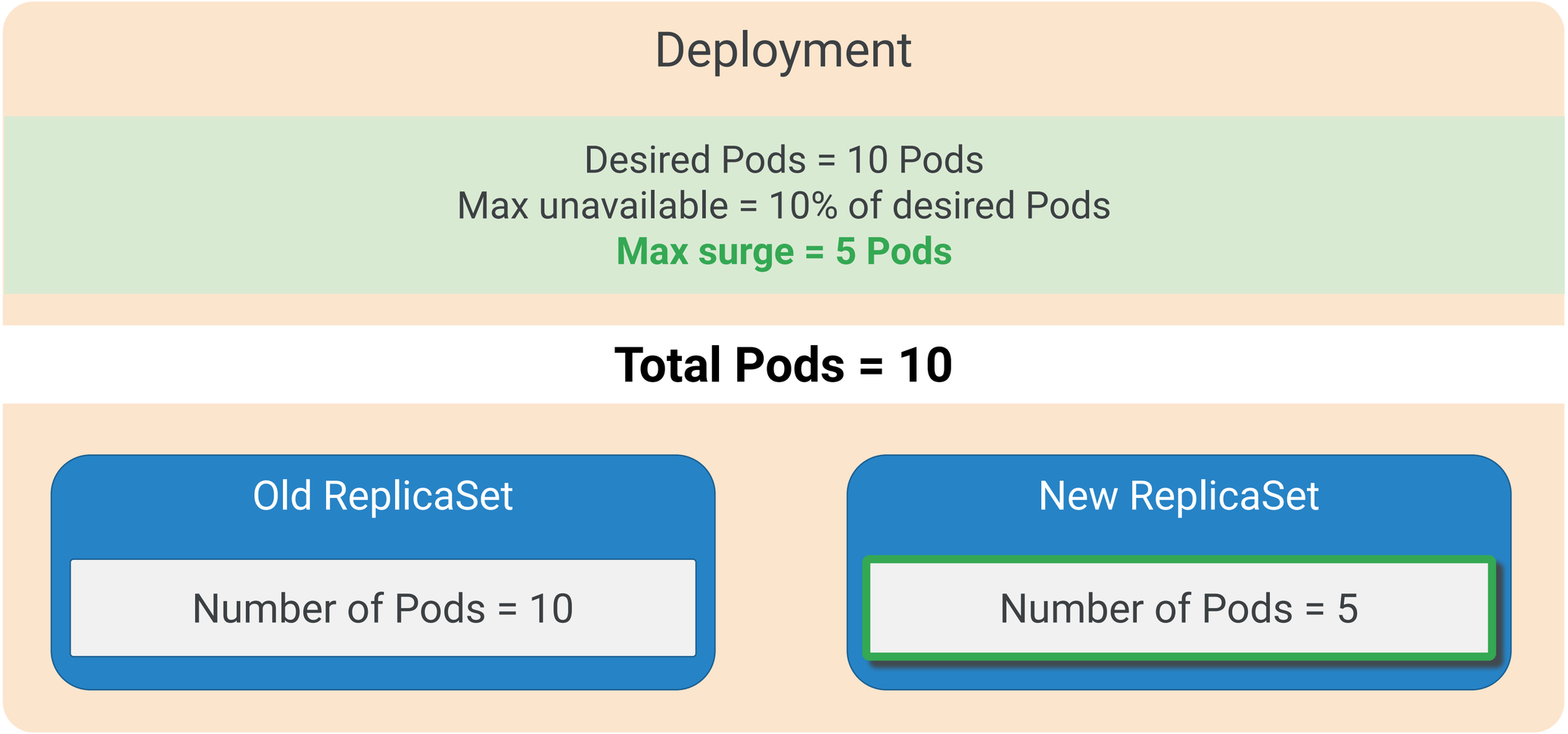
Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 10

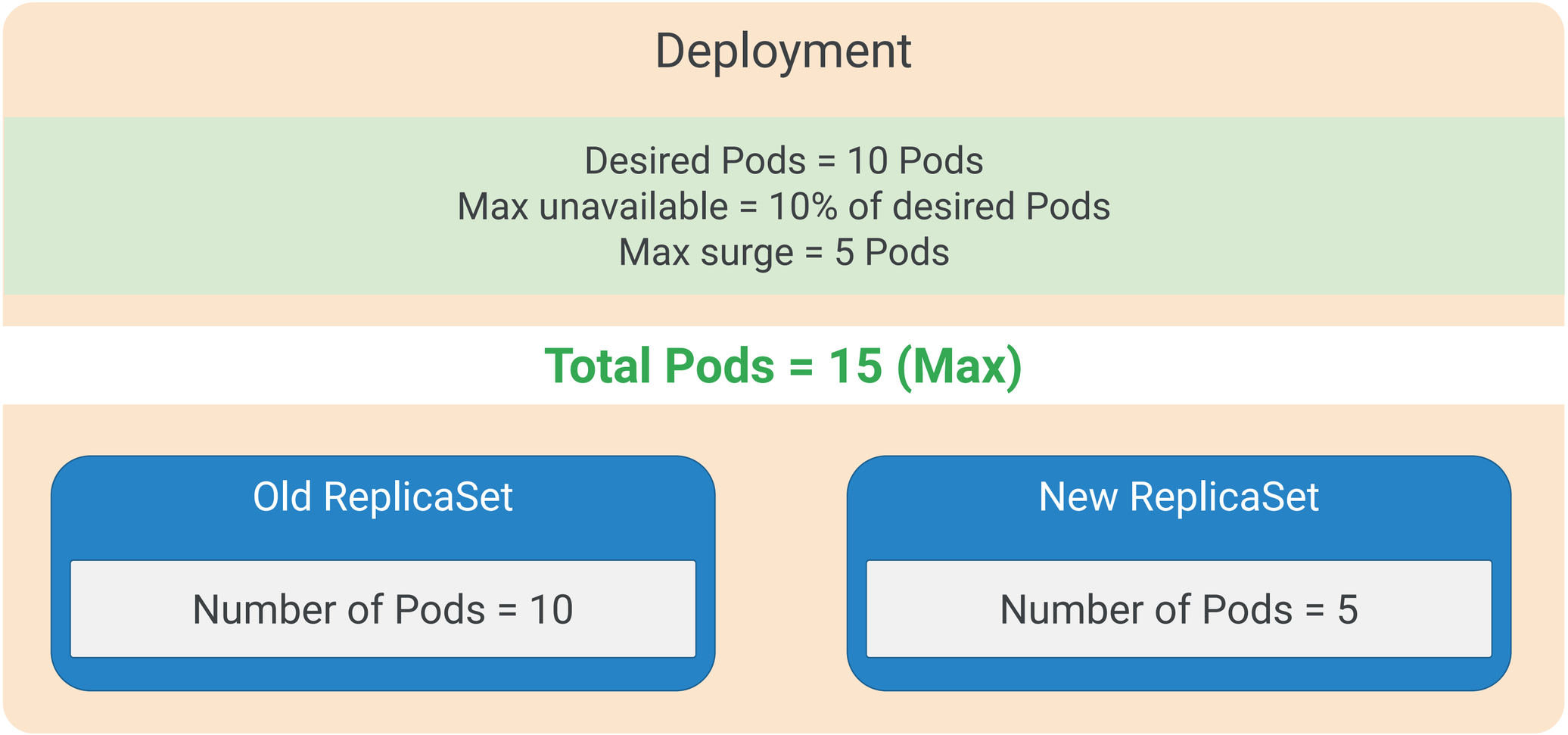
Old ReplicaSet

Number of Pods = 10

An example of a rolling update strategy



An example of a rolling update strategy



An example of a rolling update strategy

Deployment

Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 15 (Max)

Old ReplicaSet

Number of Pods = 10 **- 6** = 4

New ReplicaSet

Number of Pods = 5

An example of a rolling update strategy

Deployment

Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 9 (Min)

Old ReplicaSet

Number of Pods = 10 - 6 = **4**

New ReplicaSet

Number of Pods = **5**

An example of a rolling update strategy

Deployment

Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 9 (Min)

Old ReplicaSet

Number of Pods = 10 - 6 = 4

New ReplicaSet

Number of Pods = 5 **+ 5** = 10

An example of a rolling update strategy

Deployment

Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 14

Old ReplicaSet

Number of Pods = 10 - 6 **= 4**

New ReplicaSet

Number of Pods = 5 + 5 **= 10**

An example of a rolling update strategy

Deployment

Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 10

Old ReplicaSet

Number of Pods = 4 **- 4 = 0**

New ReplicaSet

Number of Pods = 5 + 5 = 10

An example of a rolling update strategy

Deployment

Desired Pods = 10 Pods
Max unavailable = 10% of desired Pods
Max surge = 5 Pods

Total Pods = 10

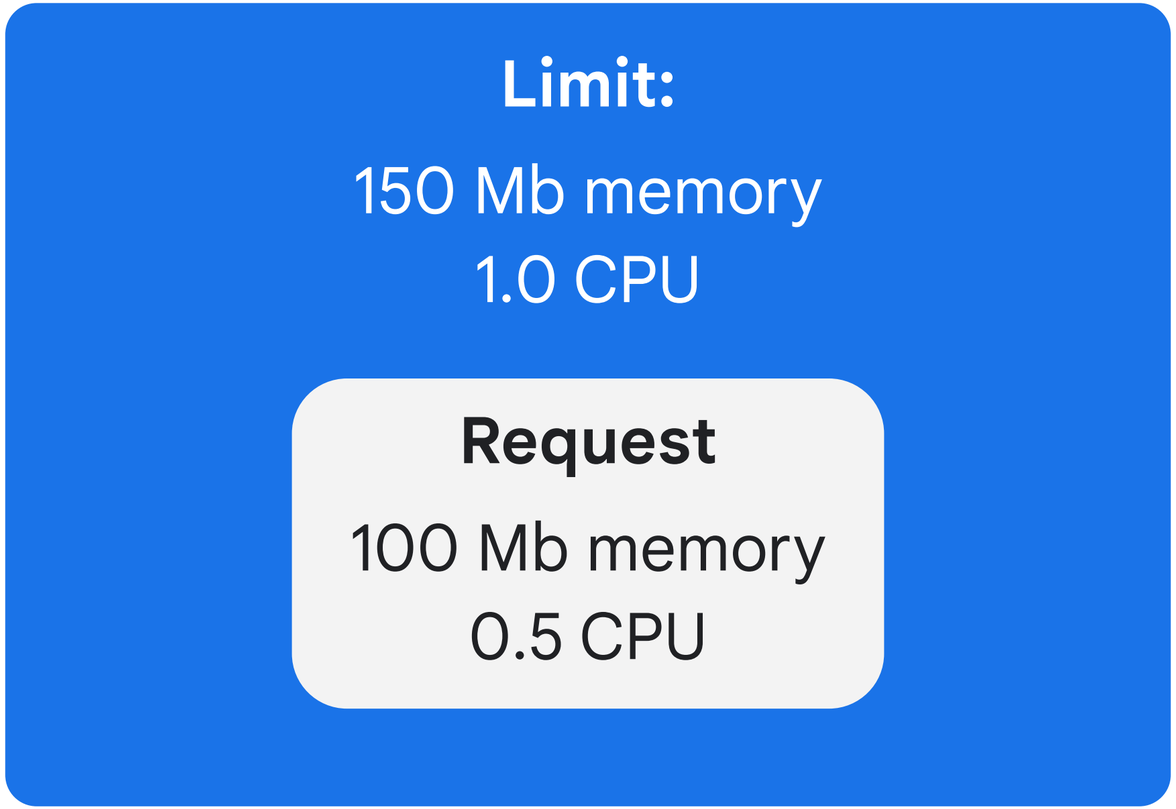
New ReplicaSet

Number of Pods = 5 + 5 = 10

Applying a Recreate strategy

```
[...]
kind: deployment
spec:
    replicas: 10
    strategy:
        type: Recreate
[...]
```

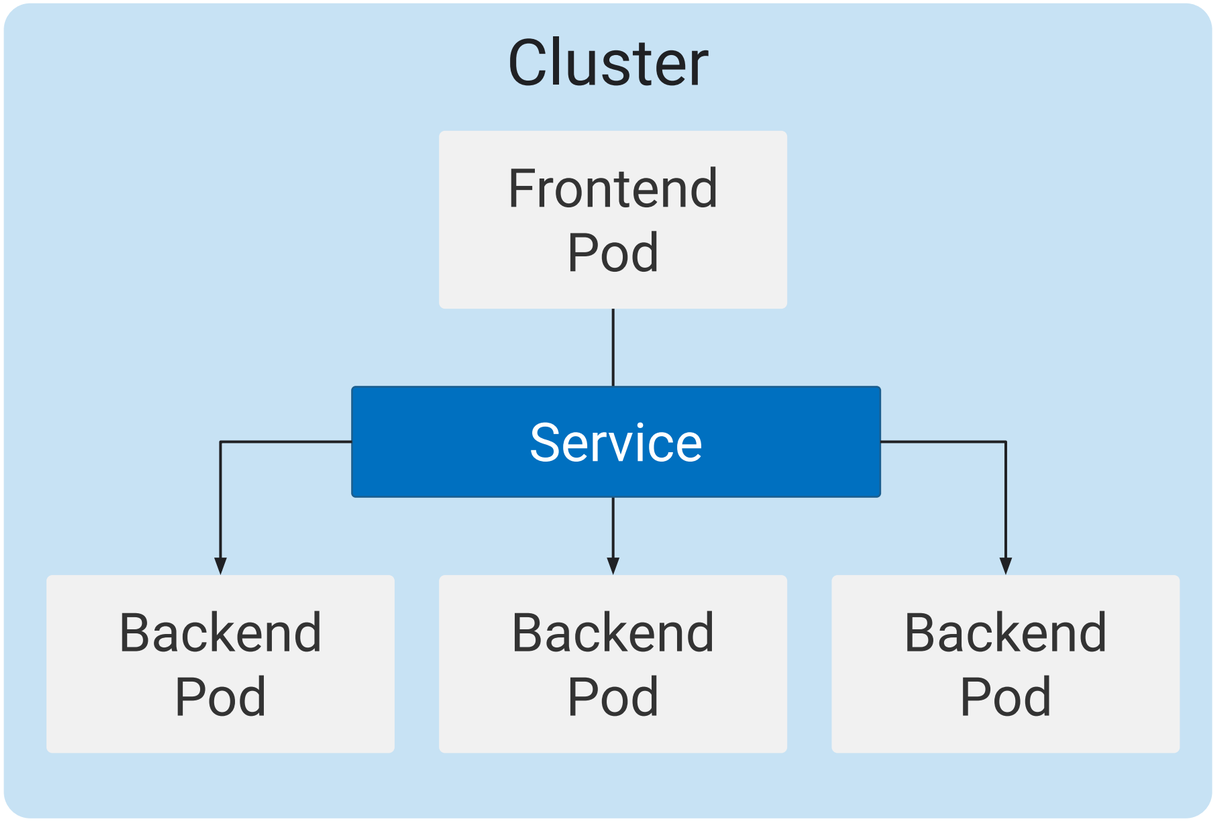
Resource requests and limits



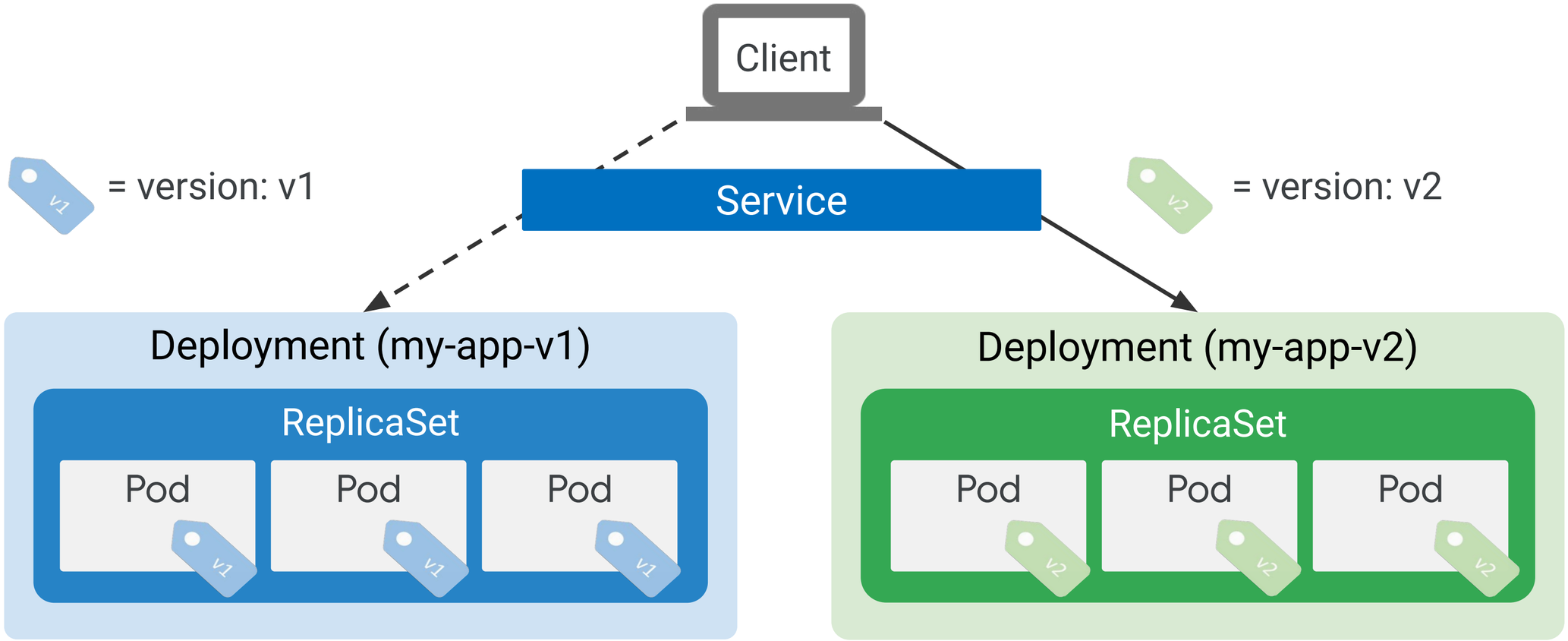
Container settings

```
containers:
- name: container1
  image: busybox
  resources:
    requests:
      memory: “32Mi”
      cpu: “250m”
    limits:
        memory: “64Mi”
      cpu: “500m”
```

Service is a stable network representation of a set of pods



A blue/green deployment strategy ensures app services remain available



Applying a blue/green deployment strategy

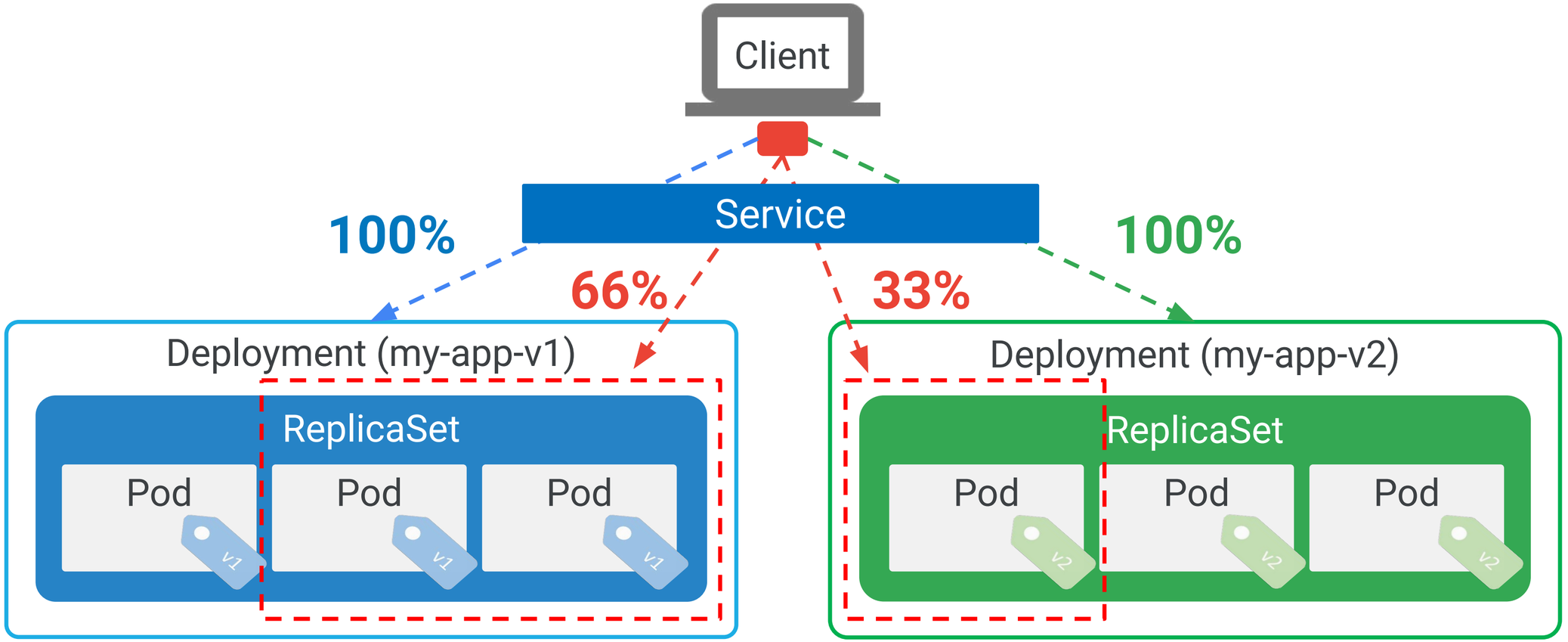
```
[...]
kind: Service
spec:
     selector:
        app: my-app
        version: v1
[...]
```

```
[...]
kind: Service
spec:
     selector:
        app: my-app
        version: v2
[...]
```

```
$ kubectl apply -f my-app-v2.yaml
```

```
$ kubectl patch service my-app-service -p \
'{"spec":{"selector":{"version":"v2"}}}
```

Canary deployment is an update strategy where traffic is gradually shifted to the new version



Applying a canary deployment

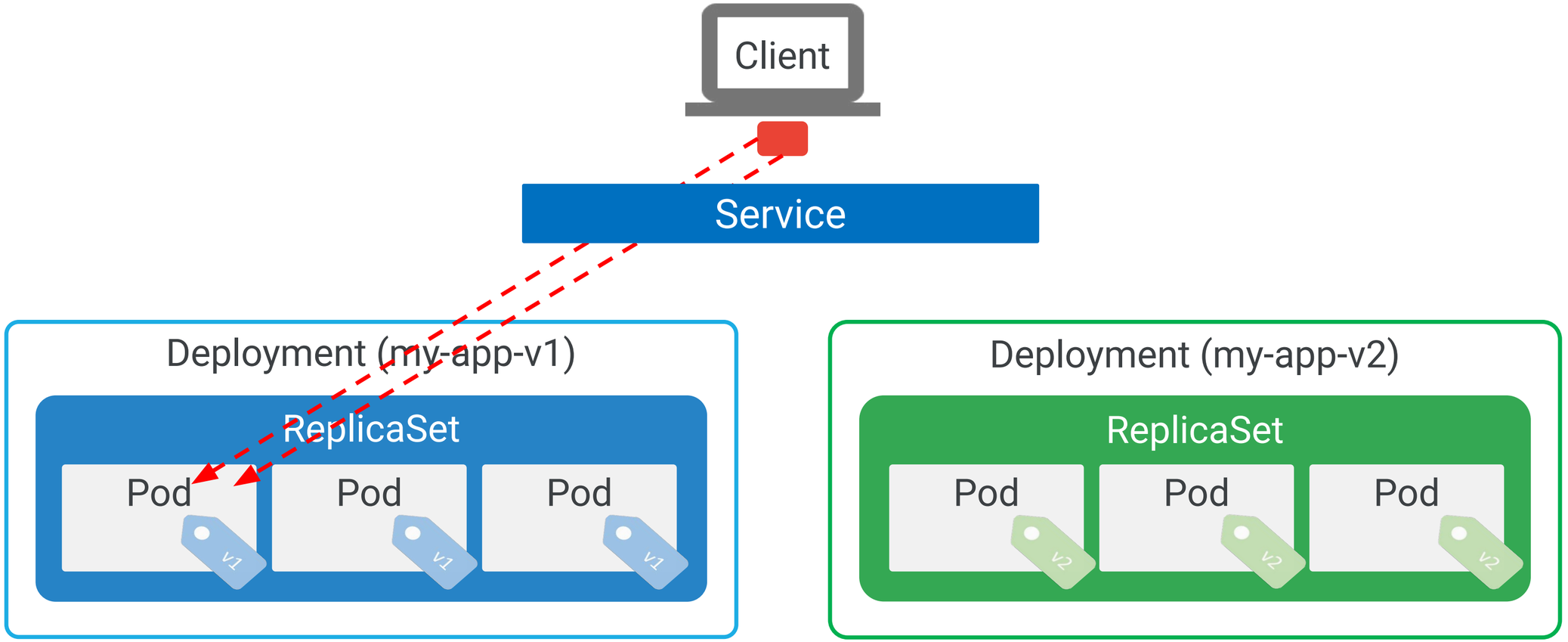
```
[...]
kind: Service
spec:
    selector:
      app: my-app
[...]
```

```
$ kubectl apply -f my-app-v2.yaml
```

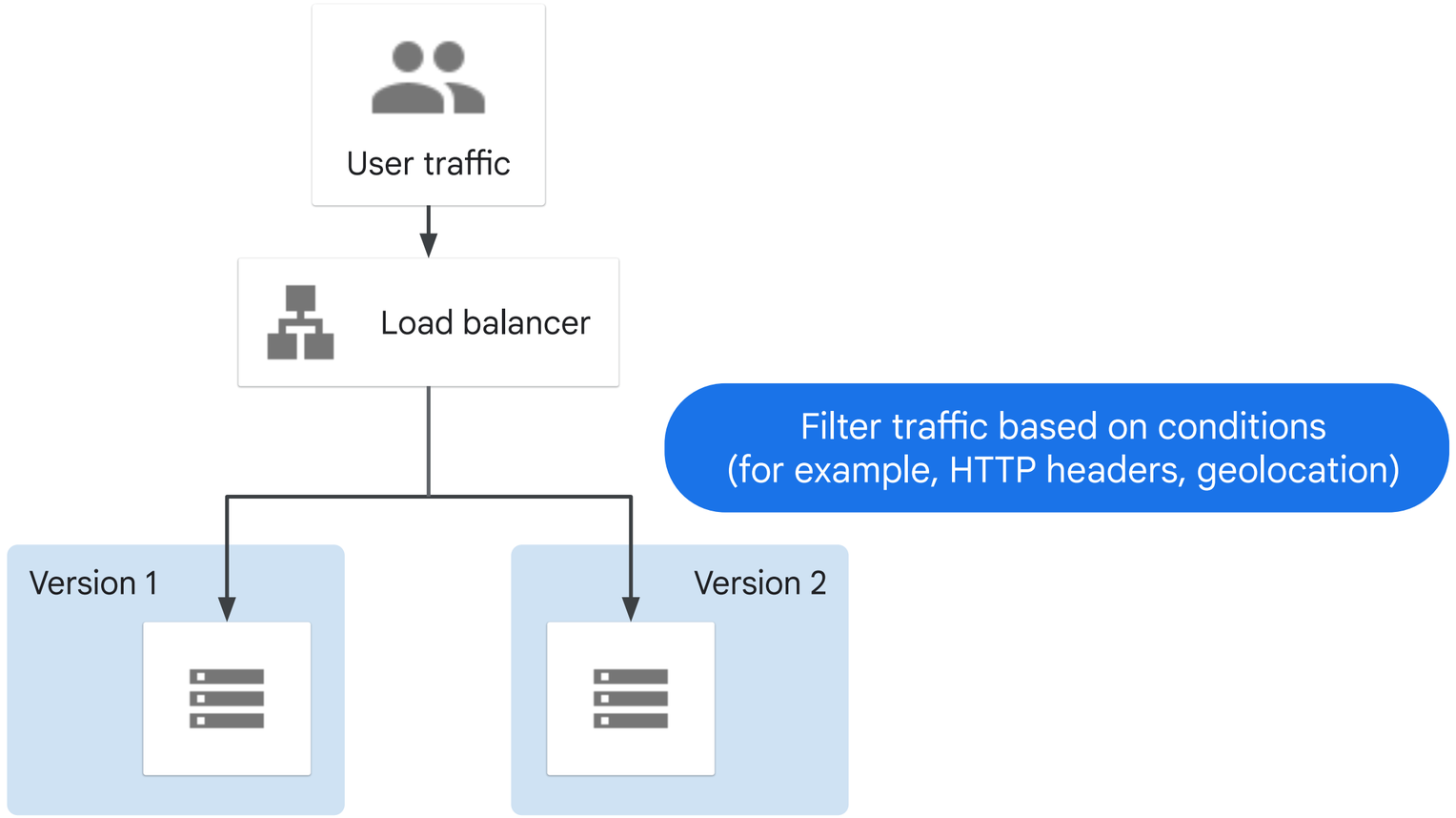
```
$ kubectl scale deploy/my-app-v2 –replicas=10
```

```
$ kubectl delete -f my-app-v1.yaml
```

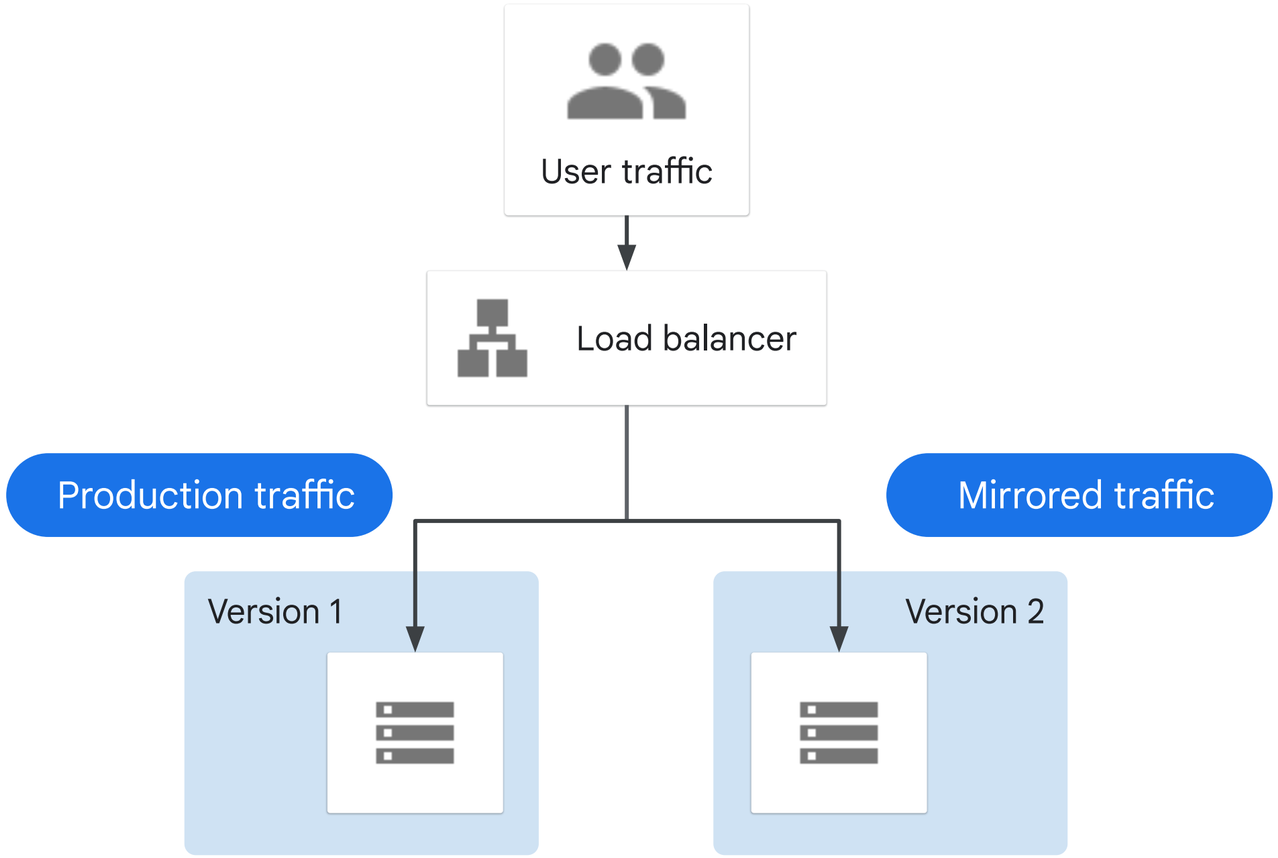
Session affinity ensures that all client requests are sent to the same Pod



A/B testing is used to measure the effectiveness of functionality in an application



Shadow testing allows you to run a new, hidden version



Choosing the right strategy

Deployment or testing pattern	Zero downtime	Real production traffic testing	Releasing to users based on conditions	Rollback duration	Impact on hardware and cloud costs
Recreate Version 1 is terminated, and Version 2 is rolled out.	✗	✗	✗	Fast but disruptive due to downtime	No extra setup required
Rolling update Version 2 is gradually rolled out and replaces Version 1.	✓	✗	✗	Slow	Can require extra setup for surge upgrades
Blue/green Version 2 is released alongside version 1; the traffic is switched to Version 2 after it is tested.	✓	✗	✗	Instant	Need to maintain blue and green environments simultaneously
Canary Version 2 is released to a subset of users, followed by a full rollout.	✓	✓	✗	Fast	No extra setup required
A/B Version 2 is released, under specific conditions, to a subset of users.	✓	✓	✓	Fast	No extra setup required
Shadow Version 2 receives real-world traffic without impacting user requests.	✓	✓	✗	Does not apply	Need to maintain parallel environments in order to capture and replay user requests

Rolling back a Deployment

```
$ kubectl rollout undo deployment [DEPLOYMENT_NAME]
```

```
$ kubectl rollout undo deployment [DEPLOYMENT_NAME] --to-revision=2
```

```
$ kubectl rollout history deployment [DEPLOYMENT_NAME] --revision=2
```

Clean up Policy:

- Default: 10 Revision
- Change: .spec.revisionHistoryLimit

Different actions can be applied to a Deployment

Pause

```
$ kubectl rollout pause deployment [DEPLOYMENT_NAME]
```

Resume

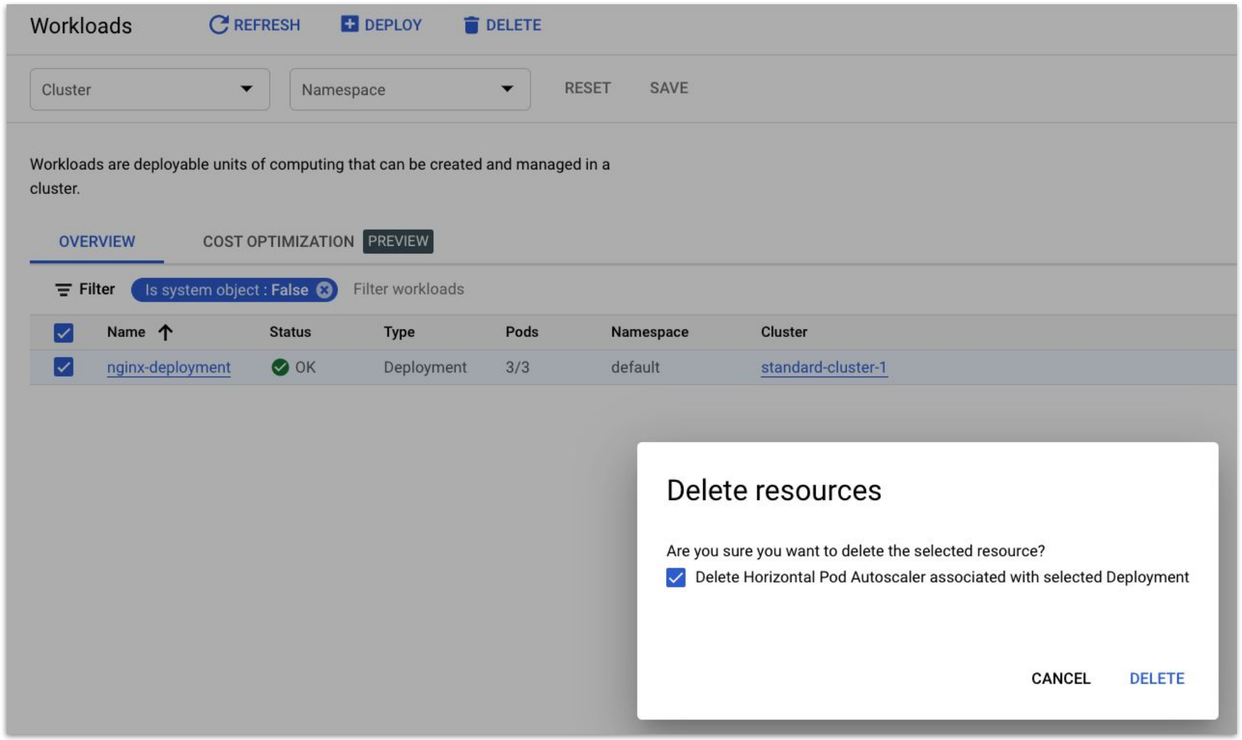
```
$ kubectl rollout resume deployment [DEPLOYMENT_NAME]
```

Monitor

```
$ kubectl rollout status deployment [DEPLOYMENT_NAME]
```

Delete a Deployment

```
$ kubectl delete deployment [DEPLOYMENT_NAME]
```



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Lab: Deploying Jobs on Google Kubernetes Engine

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Controlling Pod Placement

Getting Software into Your Cluster

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Summary

Lab Intro

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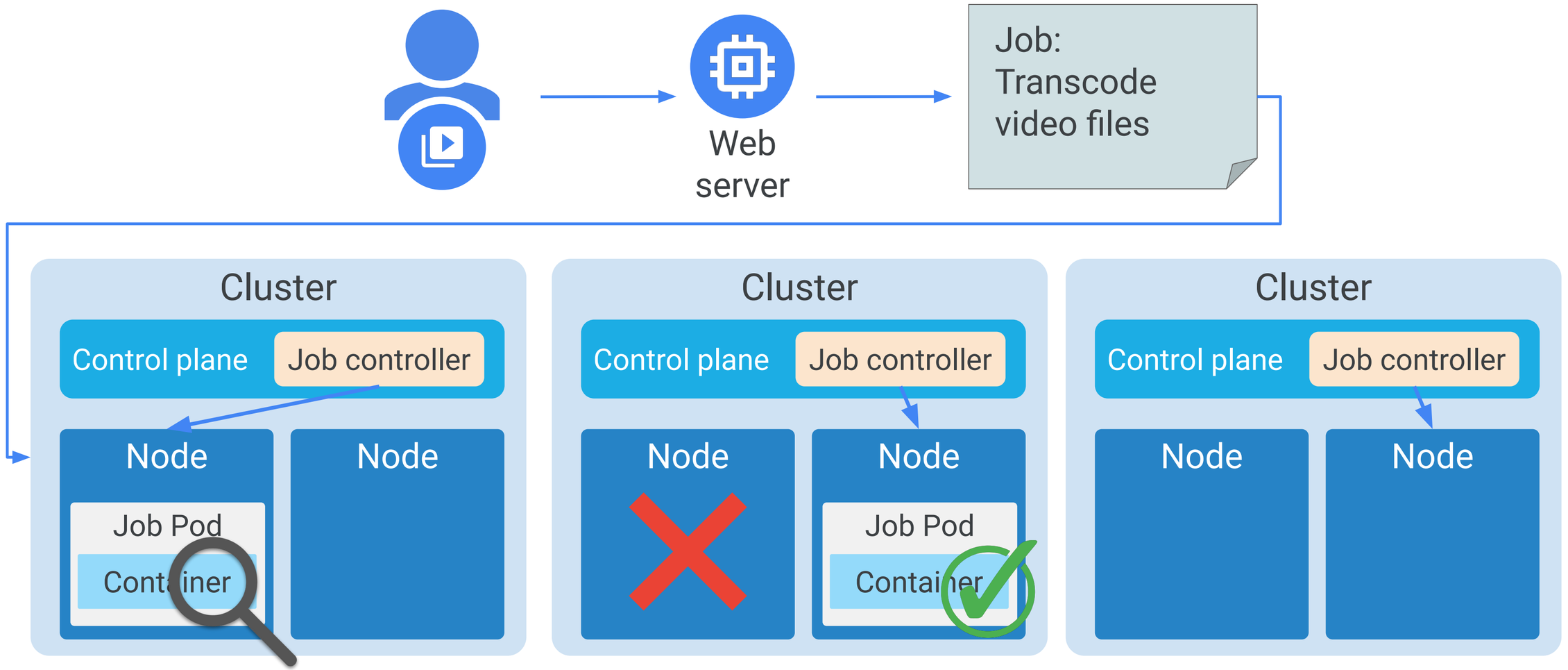
Summary

A scenario where Job provides the solution



Jobs can be defined as non-parallel or parallel

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  completions: 3
  template:
    spec:
[...]
```

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  completions: 3
  parallelism: 2
  template:
    spec:
[...]
```

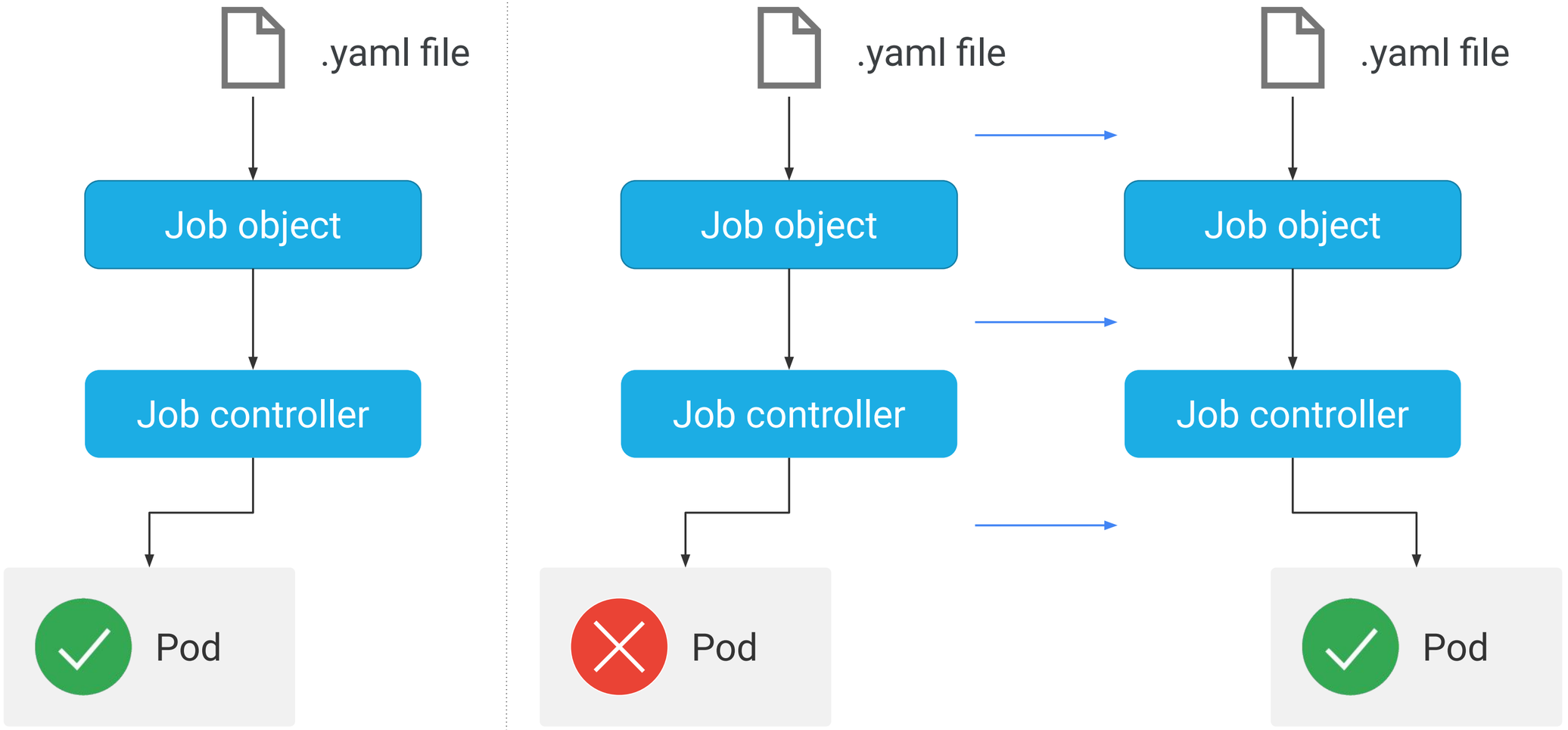
A Job computing π to 2000 places

```
apiVersion: batch/v1
kind: Job
metadata:
  name: pi
spec:
  template:
    spec:
      containers:
      - name: pi
        image: perl:5.32
        command: ["perl",  "-Mbignum=bpi", "-wle", "print bpi(2000)"]
      restartPolicy: Never
  backoffLimit: 4
```

```
$ kubectl apply -f [JOB_FILE]
```

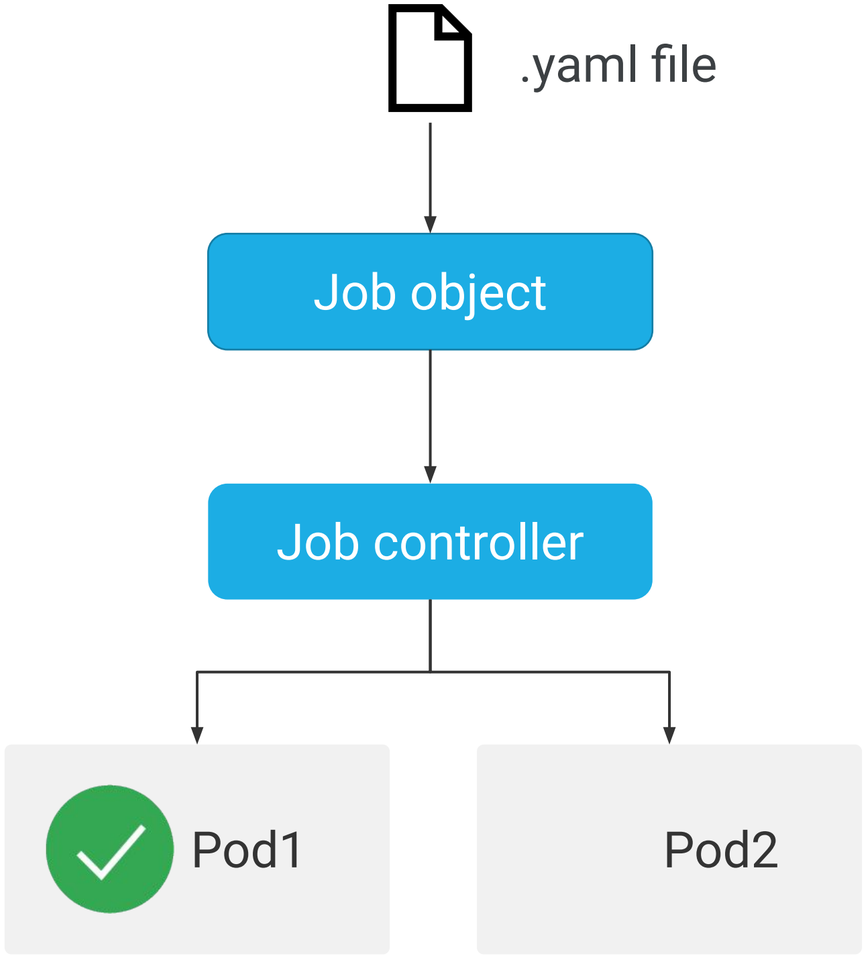
```
$ kubectl run pi --image perl:5.32 --restart Never -- perl -Mbignum=bpi -wle 'print bpi(2000)'
```

The role of a non-parallel Job



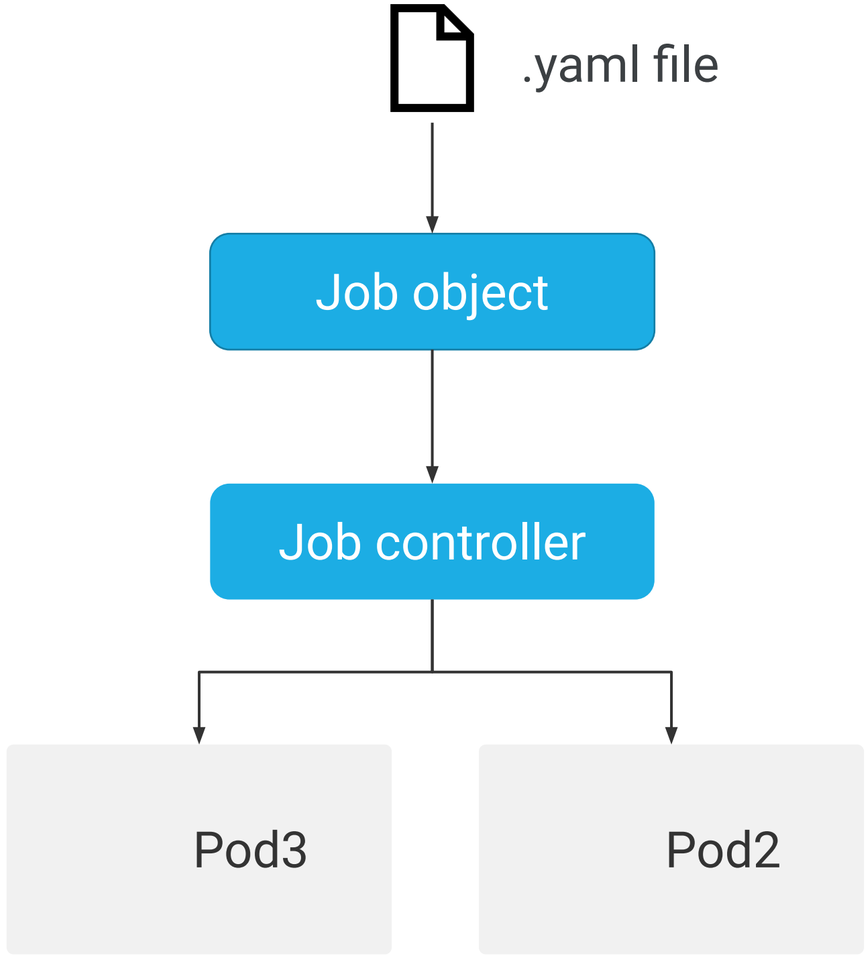
Parallel Job with fixed completion count

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  completions: 3
  parallelism: 2
  template:
    spec:
[...]
```



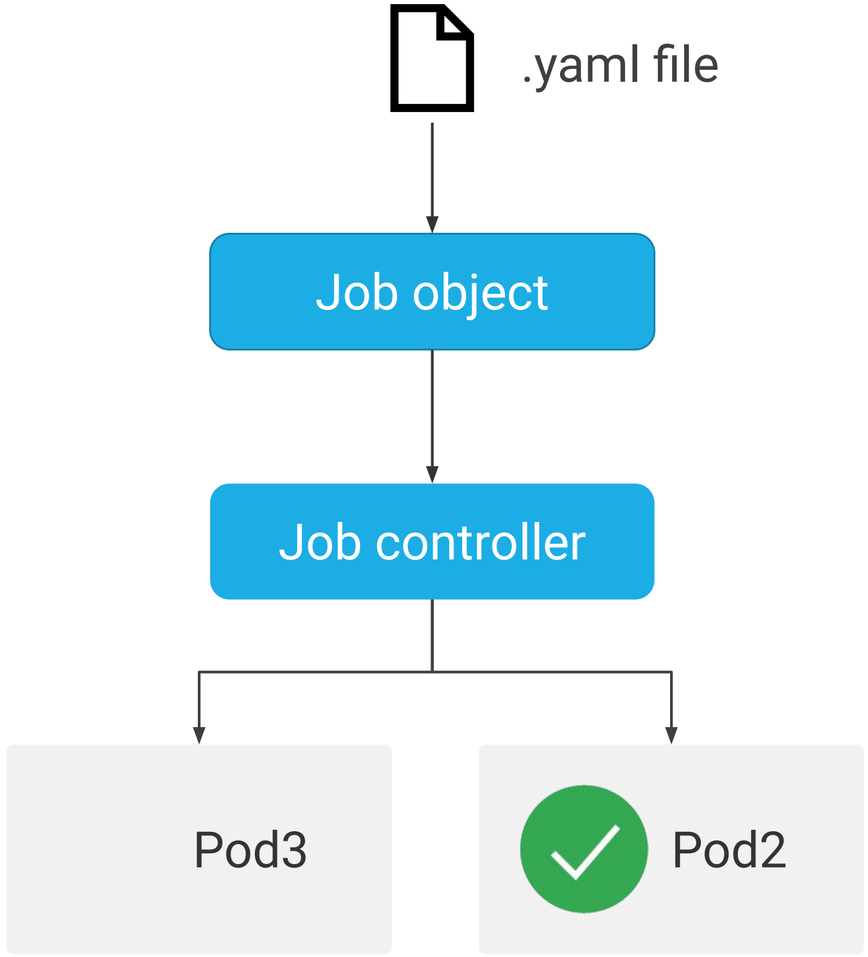
Parallel Job with fixed completion count

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  completions: 3
  parallelism: 2
  template:
    spec:
[...]
```



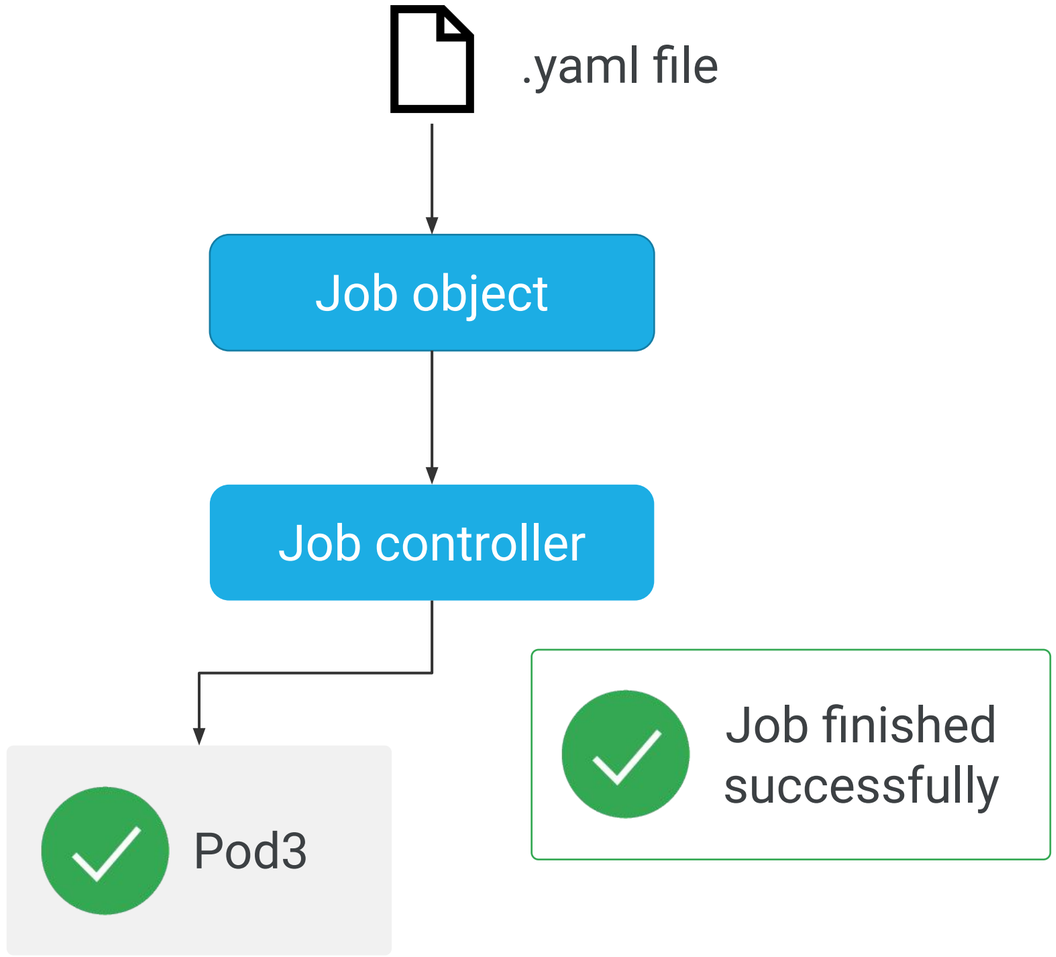
Parallel Job with fixed completion count

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  completions: 3
  parallelism: 2
  template:
    spec:
[...]
```



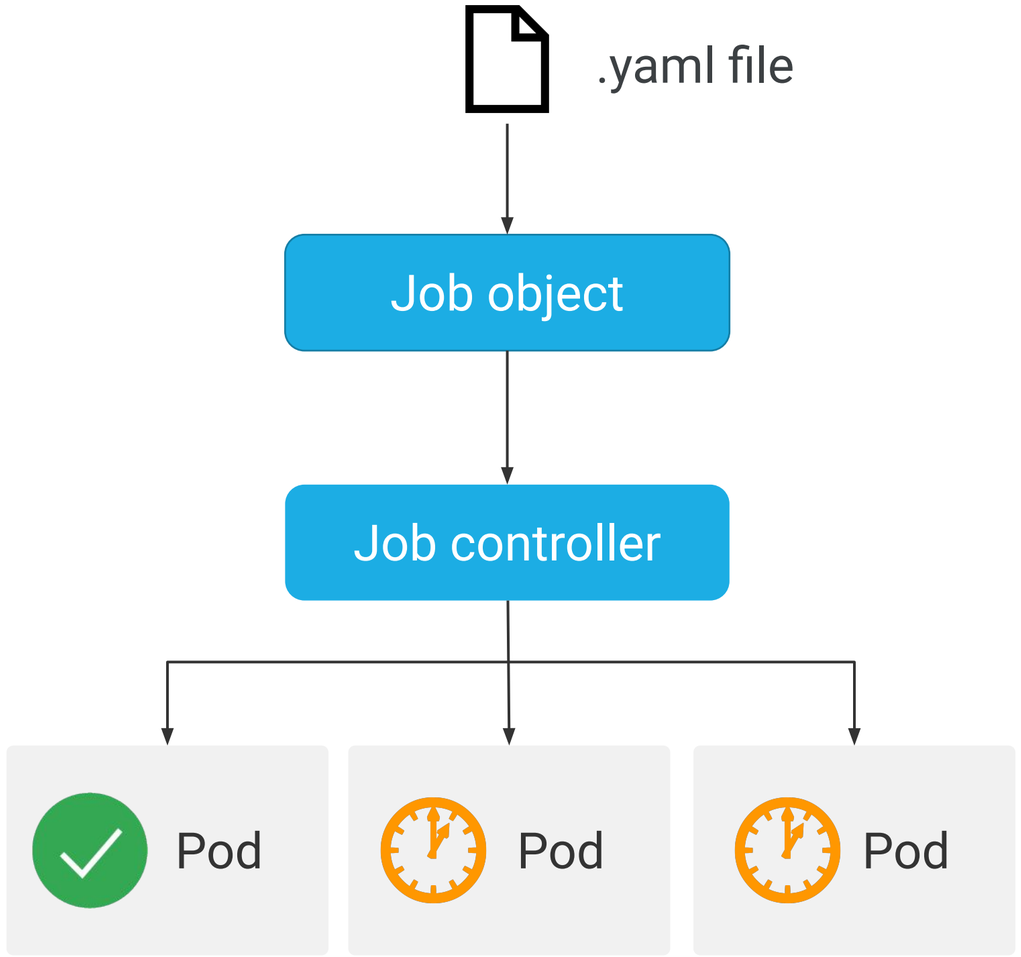
Parallel Job with fixed completion count

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  completions: 3
  parallelism: 2
  template:
    spec:
[...]
```



Parallel Job with a worker queue

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  parallelism: 3
  template:
    spec:
[...]
```



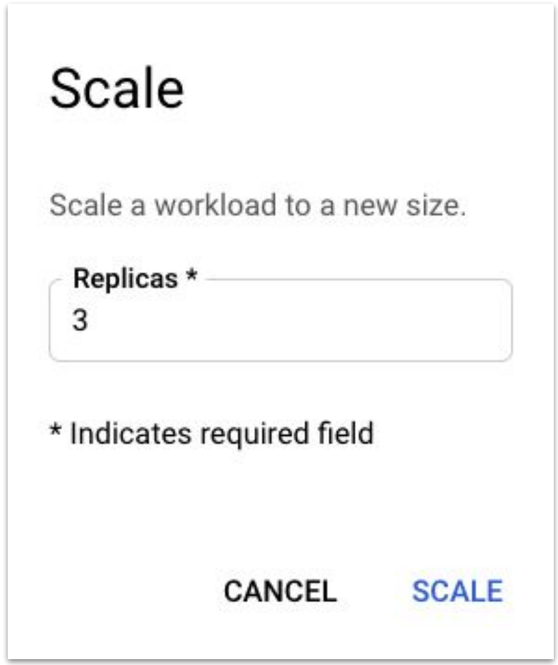
Inspecting a Job

```
$ kubectl describe job [JOB_NAME]
```

```
$ kubectl get pod -l [job-name=my-app-job]
```

Scaling a Job

```
$ kubectl scale job [JOB_NAME]
--replicas [VALUE]
```



Failing a Job

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-app-job
spec:
  backoffLimit: 4
  activeDeadlineSeconds: 300
  template:
[...]
```

Deleting a Job

```
$ kubectl delete -f [JOB_FILE]
```

```
$ kubectl delete job [JOB_NAME]
```

```
$ kubectl delete job [JOB_NAME]
  --cascade false
```

Cron format is a simple, yet powerful and flexible way to define a schedule

```
Min   Hour Day   Mon   Weekday

*     *    *     *     *  command to be executed

┬     ┬    ┬     ┬     ┬
│     │    │     │     └─  Weekday  (0=Sun .. 6=Sat)
│     │    │     └───────  Month    (1..12)
│     │    └─────────────  Day      (1..31)
│     └──────────────────  Hour     (0..23)
└────────────────────────  Minute   (0..59)
```

Examples	
`0 * * * *`	Every hour
`*/15 * * * *`	Every 15 minutes
`0 */6 * * *`	Every 6 hours
`0 20 * * 1-5`	Every week Mon-Fri at 8pm
`0 0 1 */2 *`	At 00:00 on day 1 of every other month

Setting up a cron schedule under the CronJob spec

```
apiVersion: batch/v1
kind: CronJob
metadata:
   name: my-app-job
spec:
   schedule: "*/1 * * * *"
   jobtemplate:
      spec:
         template:
            spec:
[...]
```

Setting up a cron schedule under the CronJob spec

```
apiVersion: batch/v1
kind: CronJob
metadata:
   name: my-app-job
spec:
   schedule: "*/1 * * * *"
   startingDeadlineSeconds: 100
   jobtemplate:
      spec:
         template:
            spec:
[...]
```

Setting up a cron schedule under the CronJob spec

```
apiVersion: batch/v1
kind: CronJob
metadata:
   name: my-app-job
spec:
   schedule: "*/1 * * * *"
   startingDeadlineSeconds: 100
   concurrencyPolicy: Forbid
   jobtemplate:
      spec:
         template:
            spec:
[...]
```

Setting up a cron schedule under the CronJob spec

```
apiVersion: batch/v1
kind: CronJob
metadata:
   name: my-app-job
spec:
   schedule: "*/1 * * * *"
   startingDeadlineSeconds: 100
   concurrencyPolicy: Forbid
   suspend: True
   successfulJobsHistoryLimit: 3
   failedJobsHistoryLimit: 1
   jobtemplate:
      spec:
[...]l
```

CronJobs can be managed using kubectl

Create a CronJob

```
$ kubectl apply -f [FILE]
```

Inspect a CronJob

```
$ kubectl describe cronjob [NAME]
```

Delete a CronJob

```
$ kubectl delete cronjob [NAME]
```

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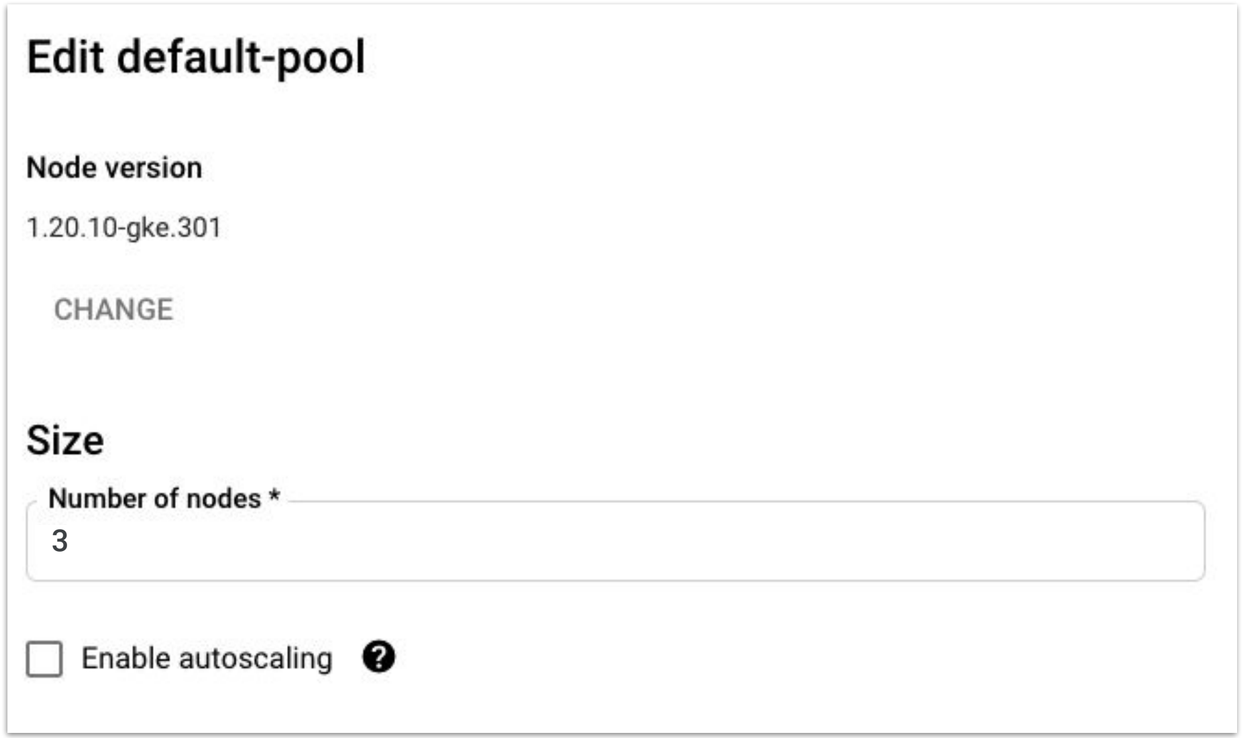
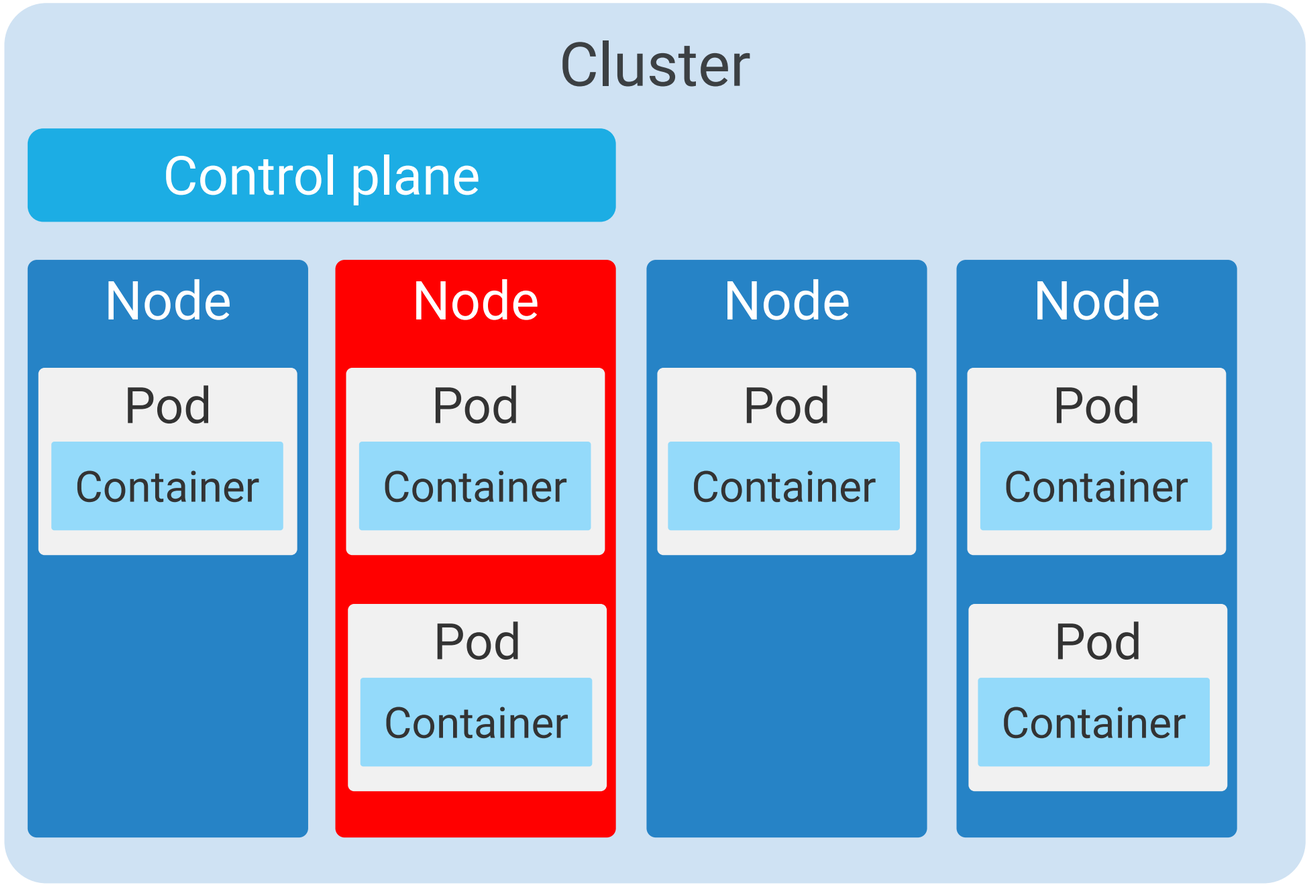
Scaling down a cluster using the gcloud command

```
gcloud container clusters resize
projectdemo --node-pool default-pool \
--num_nodes 6
```

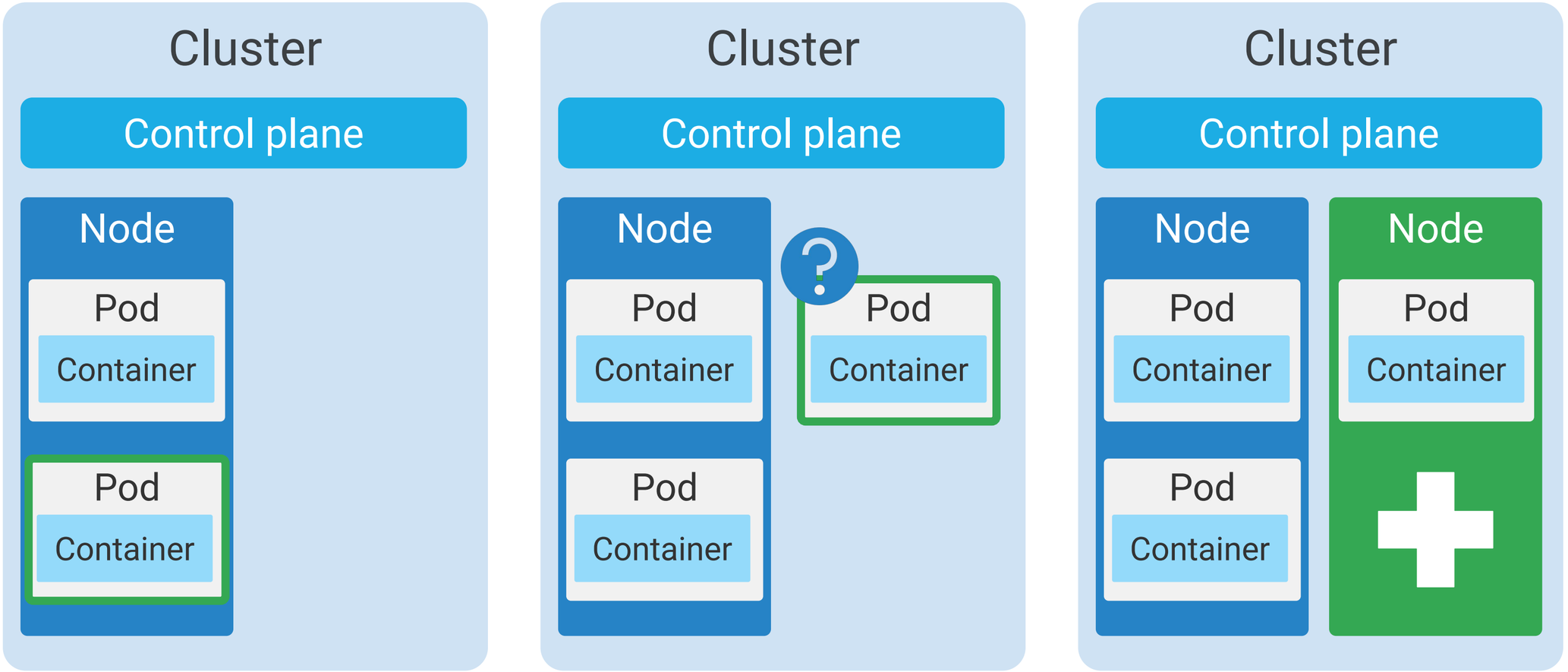
Scaling a cluster from the Cloud Console



Manual Cluster Scale down selects nodes randomly



Scale up a cluster with autoscaling



Scale down a cluster with autoscaling

- There can be no scale-up events pending.
- Can the node be deleted safely?

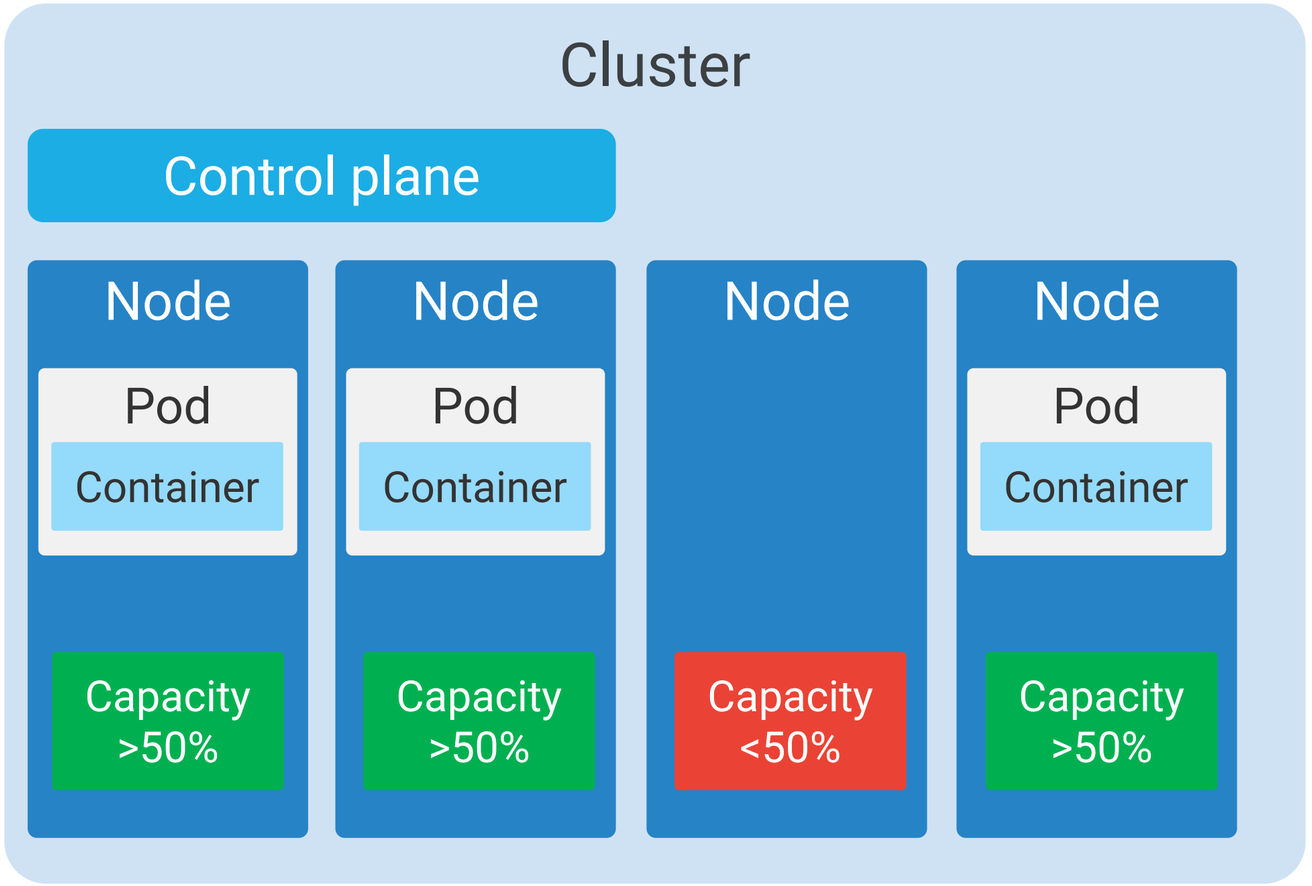
Pod conditions that prevent node deletion

- Not run by a controller.
- Has local storage.
- Restricted by constraint rules.

Pod conditions that prevent node deletion

- `cluster-autoscaler.kubernetes.io/safe-to-evict` is set to False
- Restrictive PodDisruptionBudget
- `kubernetes.io/scale-down-disabled` set to True

The Autoscaler removes nodes that remain below 50% utilization



Best practices for working with autoscaled clusters

Don’t run Compute Engine autoscaling

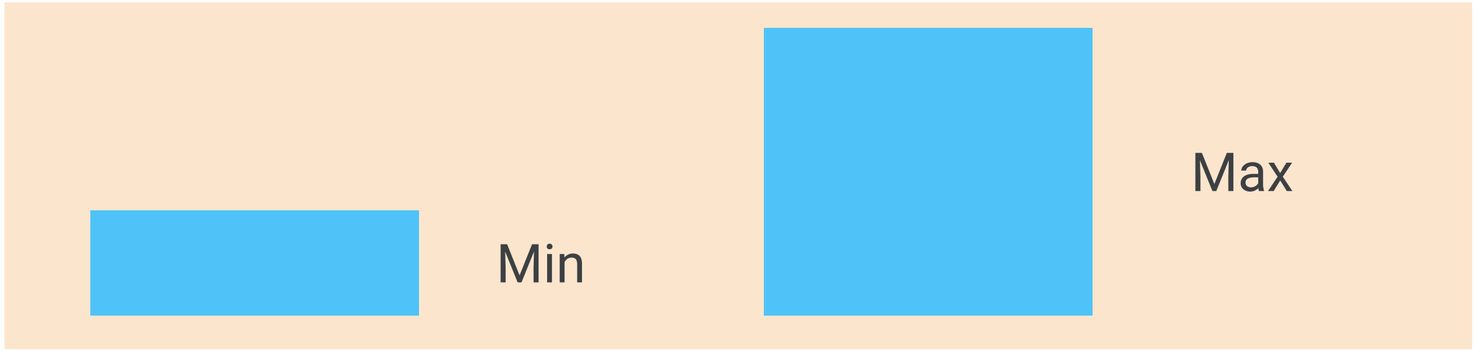
Don’t manually resize a node using the gcloud command

Don’t manually modify a node

Specify correct resource requests for Pods

Use PodDisruptionBudget to maintain the app’s availability

Setting a node pool size



Node pool = 0

Cluster size ≠ 0

MAX = 15,000 nodes x 110 Pods

Increase quota limits to avoid disruption

gcloud commands for autoscaling

Create a cluster with autoscaling enabled

```
gcloud container clusters create [CLUSTER_NAME] --enable-autoscaling --min-nodes 15 --max-nodes 50 [--zone COMPUTE_ZONE]
```

Enable autoscaling for an existing node pool

```
gcloud container clusters update [CLUSTER_NAME] --enable-autoscaling \
--min-nodes 1 --max-nodes 10 --zone [COMPUTE_ZONE] --node-pool [POOL_NAME]
```

Add a node pool with autoscaling enabled

```
gcloud container node-pools create [POOL_NAME] --cluster [CLUSTER_NAME] --enable-autoscaling --min-nodes 15 --max-nodes 50 [--zone COMPUTE_ZONE]
```

Disable autoscaling for an existing node pool

```
gcloud container clusters update [CLUSTER_NAME] --no-enable-autoscaling \
--node-pool [POOL_NAME] [--zone [COMPUTE_ZONE] --project [PROJECT_ID]]
```

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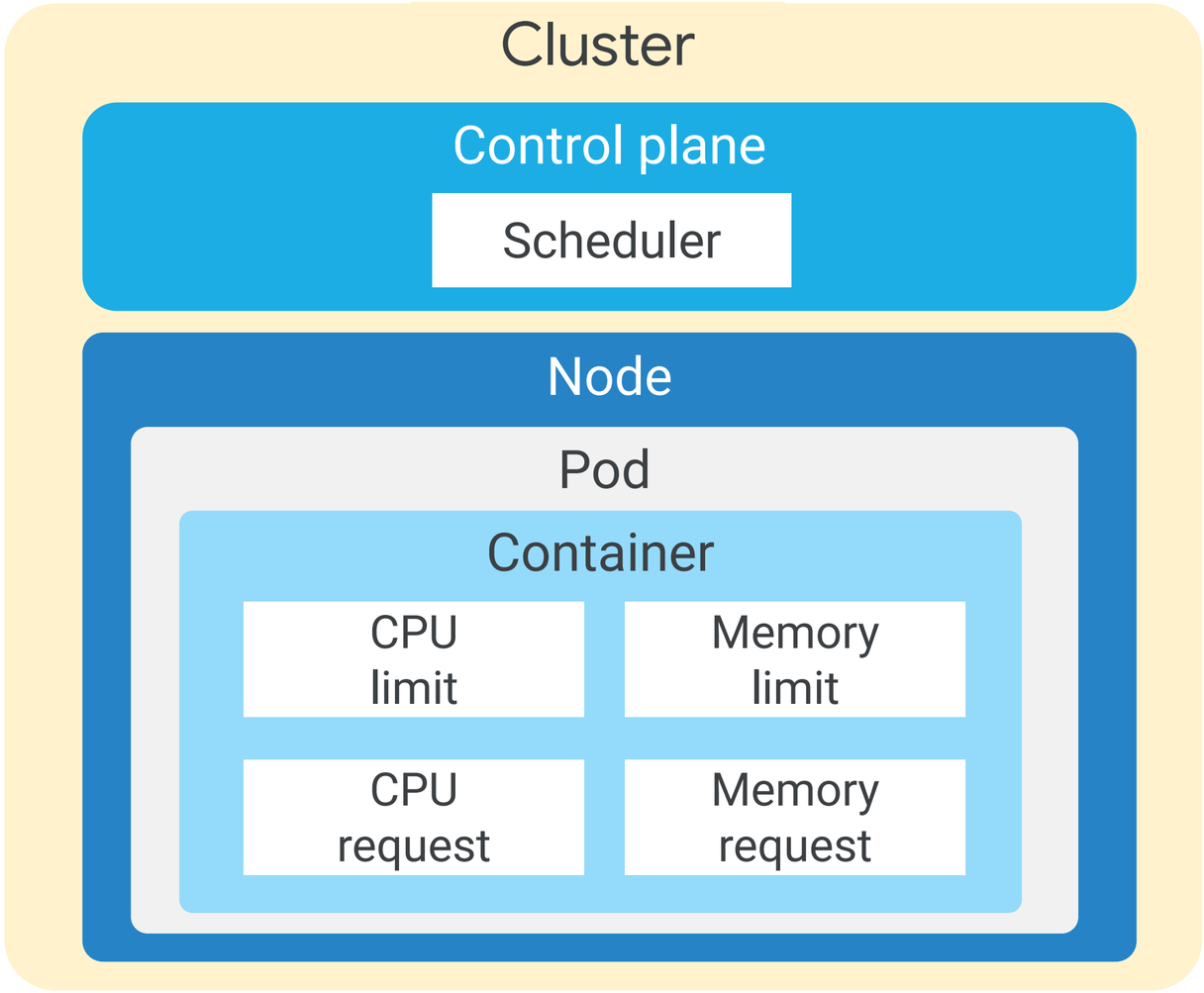
Controlled scheduling



Nodes must match all the labels present under the nodeSelector field

```
apiVersion: v1
kind: Pod
metadata:
  name: mysql
  labels:
    env: test
spec:
  containers:
  - name: mysql
    image: mysql
    imagePullPolicy: IfNotPresent
  nodeSelector:
    disktype: ssd
[…]
```

```
apiVersion: v1
kind: Node
metadata:
  name: node1
  labels:
    disktype: ssd
[…]
```

Nodes must match all the labels present under the nodeSelector field

```
apiVersion: v1
kind: Pod
metadata:
  name: mysql
  labels:
    env: test
spec:
  containers:
  - name: mysql
    image: mysql
    imagePullPolicy: IfNotPresent
  nodeSelector:
    cloud.google.com/gke-nodepool=ssd
[...]
```

Node affinity is conceptually similar to nodeSelector

```
apiVersion: v1
kind: Pod
metadata:
  name: with-node-affinity
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
        - matchExpressions:
          - key: beta.kubernetes.io/instance-type
            operator: In
            values:
            - n1-highmem-4
            - n1-highmem-8
[...]
```

Affinity and anti-affinity rules

```
apiVersion: v1
kind: Pod
metadata:
  name: with-node-affinity
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
        - matchExpressions:
          - key: accelerator-type
            operator: In
            values:
            - gpu
            - tpu
```

Defining the intensity of preference

```
apiVersion: v1
kind: Pod
metadata:
  name: with-node-affinity
spec:
  affinity:
    nodeAffinity:
      preferredDuringSchedulingIgnoredDuringExecution:
        - weight: 90
          preference:
            - matchExpressions:
              - key: cloud.google.com/gke-nodepool
                operator: In
                values:
                - n1-highmem-4
                - n1-highmem-8
```

Inter-pod affinity and anti-affinity features are built on the node affinity concept

```
[...]
metadata:
  name: with-pod-affinity
spec:
  affinity:
    podAntiAffinity:
      preferredDuringSchedulingIgnoredDuringExecution:
      - weight: 100
        podAffinityTerm:
         labelSelector:
           matchExpressions:
           - key: app
             operator: In
             values:
             - webserver
         topologyKey: failure-domain.beta.kubernetes.io/zone
```

Combining inter-pod affinity and anti-affinity

Pod (#1)
(Label – app:webserver)

Requirement: **No app:webserver**
on the **same zone**

Preference: **Prefer app:cache**
on the **same node**

Pod (#2)
(Label – app:webserver)

Requirement: No app:webserver
on the same zone

Preference:Prefer app:cache
on the same node

Pod (#3)
(Label – app:cache)

Requirement: No app:cache
on the **same zone**

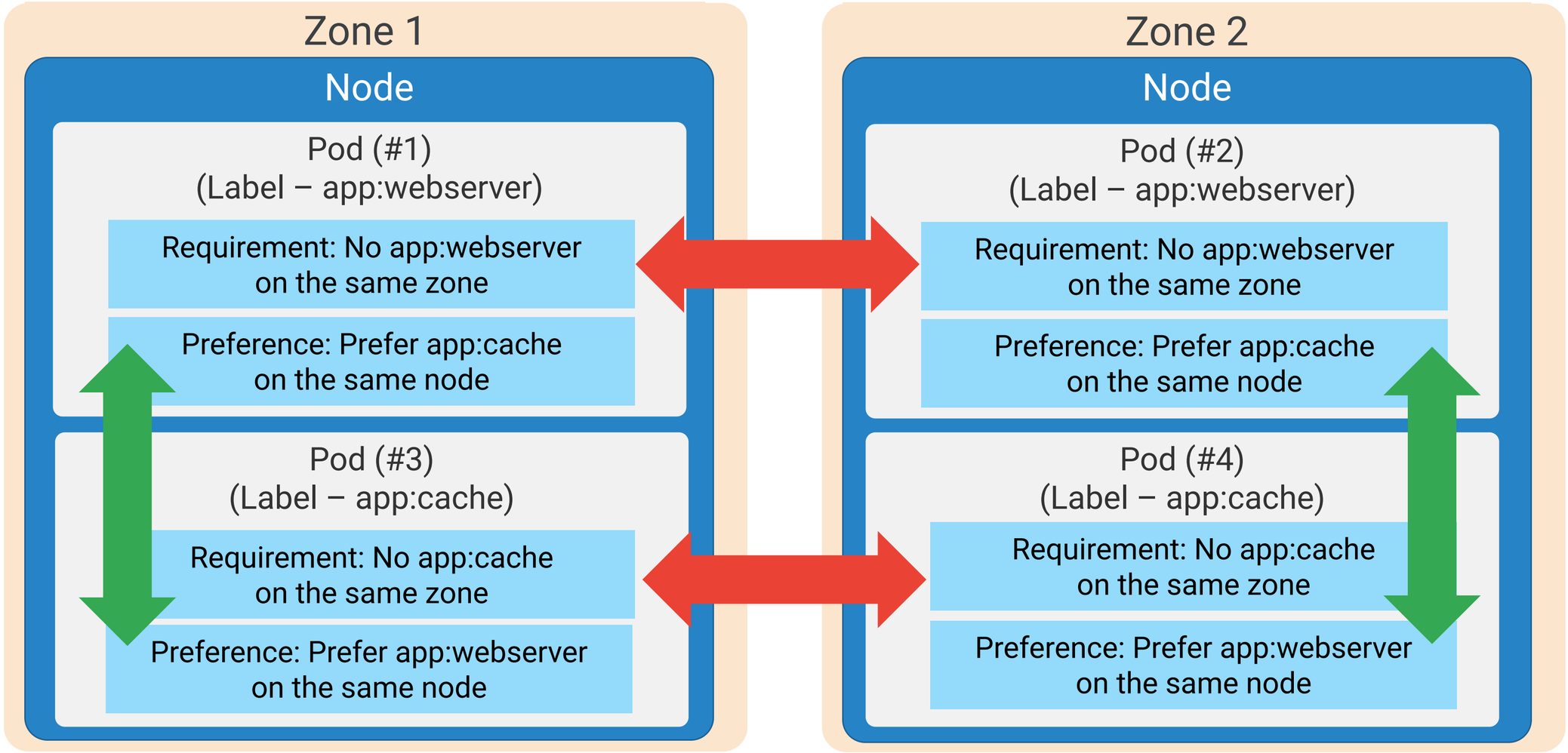
Preference: Prefer app:webserver
on the **same node**

Pod (#4)
(Label – app:cache)

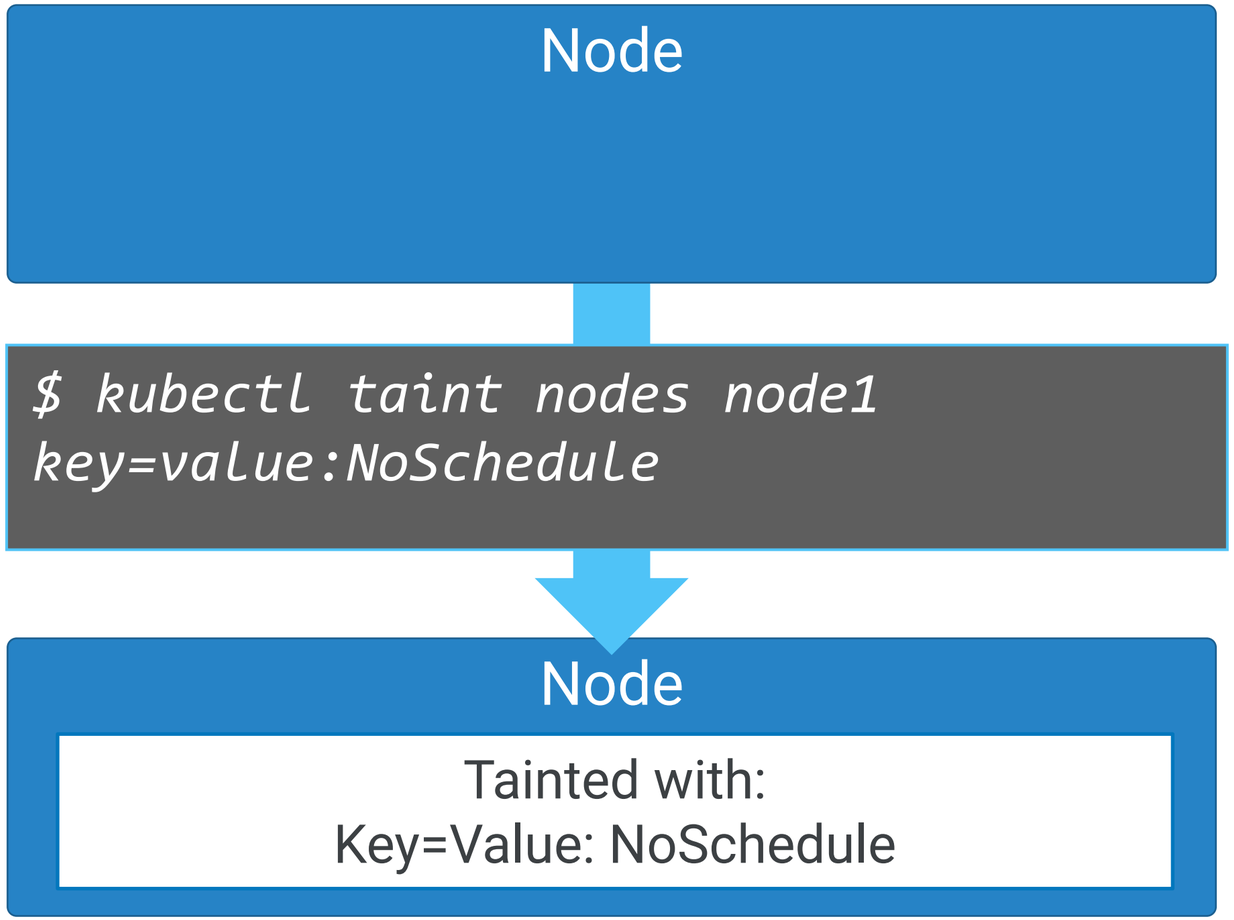
Requirement No app:cache
on the same zone

Preference: Prefer app:webserver
on the same node

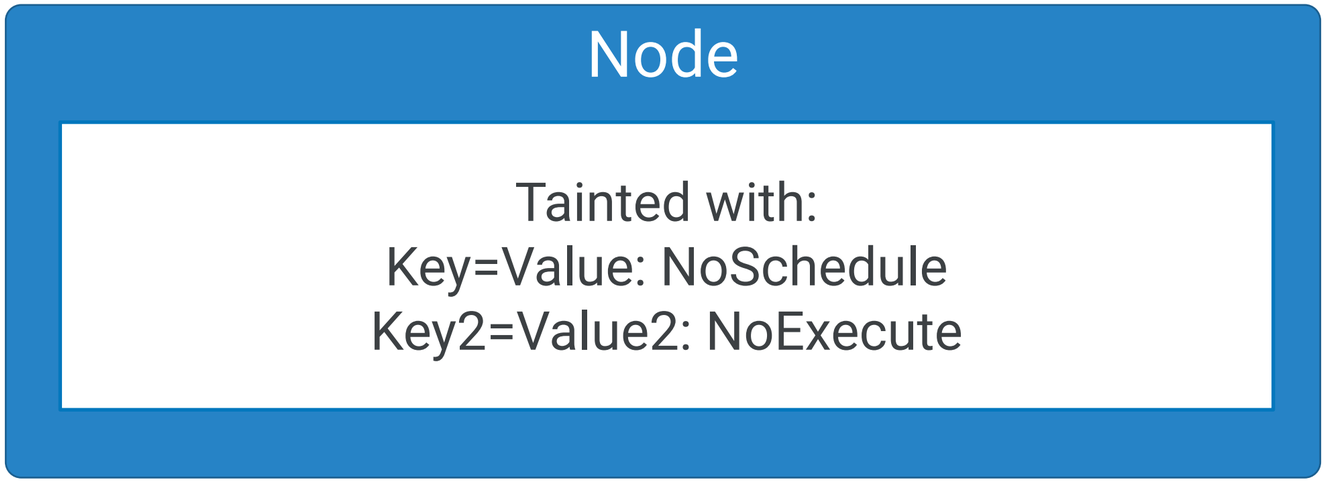
Combining inter-pod affinity and anti-affinity



Taints allow a node to repel Pods



You can apply multiple taints to a node



You can configure Pods to tolerate taints

Node

Tainted with:
Key=Value: NoSchedule

Pod

```
tolerations:
- key: "key"
  operator: "Equal"
  value: "value"
  effect: "NoSchedule"
```

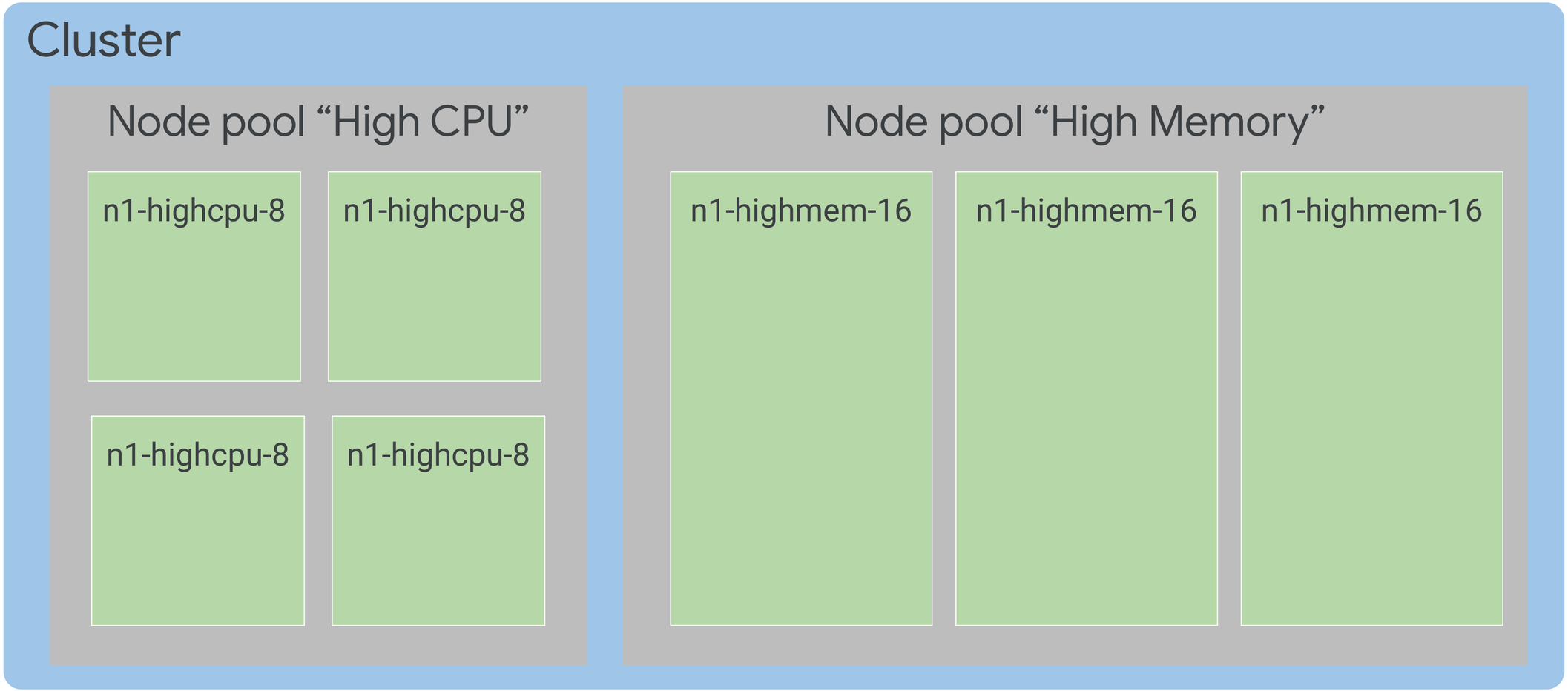
Pod

```
tolerations:
- key: "key"
  operator: "Exists"
  effect: "NoSchedule"
```

Pod

```
tolerations:
- key: "key"
  operator: "Exists"
  effect:
"PreferNoSchedule"
```

Use node pools to manage different kinds of nodes



Direct Pods to desired nodes using nodeSelectors

```
apiVersion: v1
kind: Pod
metadata:
   name: nginx
spec:
  containers:
  - name: nginx
    image: nginx
  nodeSelector:
    cloud.google.com/gke-nodepool: HighCPU
[...]
```

Or use node pool names with affinity, anti-affinity rules

```
apiVersion: v1
kind: Pod
metadata:
  name: rendering-engine
spec:
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
            - key: cloud.google.com/gke-nodepool
              operator: NotIn
              values:
               - HighCPU
```

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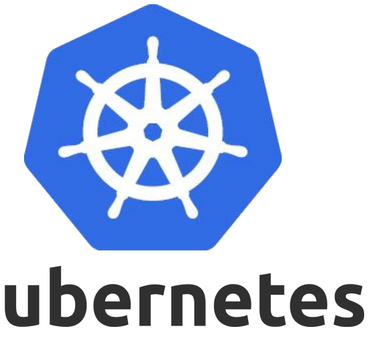
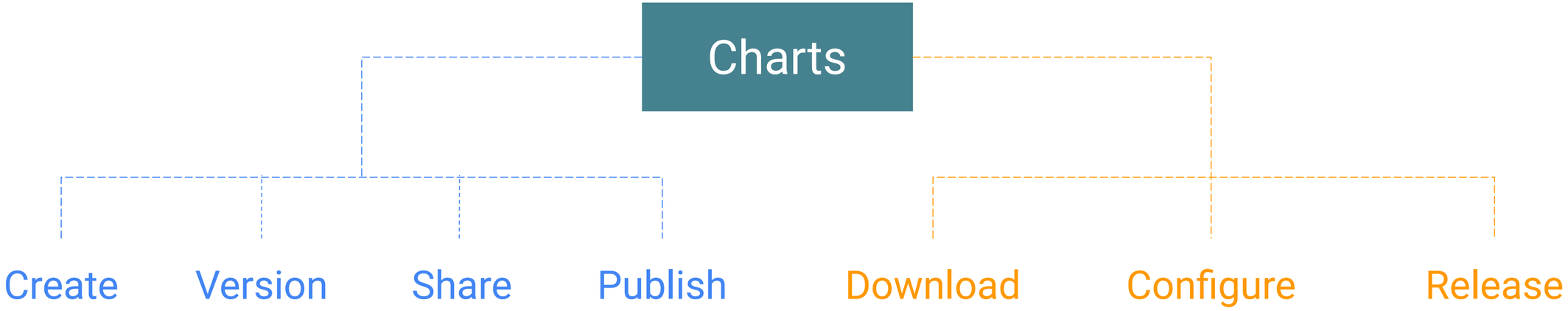
How to get software

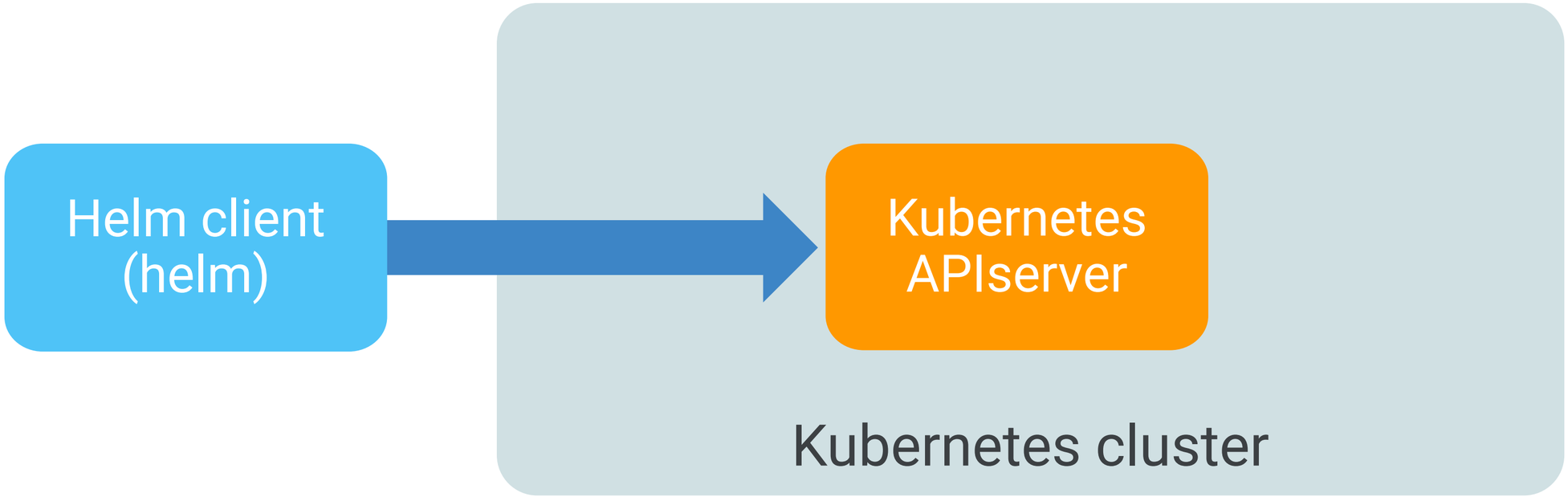
- Build it yourself, and supply your own YAML.
- Use Helm to install software into your cluster.

Organize Kubernetes objects in packages and deploy complex packages





Helm interacts directly with the Kubernetes APIserver



How to get software

- Build it yourself, and supply your own YAML.
- Use Helm to install software into your cluster.
- Use Google Cloud Marketplace to install both open-source and commercial software.

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Create and use Deployments.

Create and run Jobs and CronJobs.

Use Helm Charts.

Scale clusters manually and automatically.

Configure Node and Pod affinity.


Google Cloud